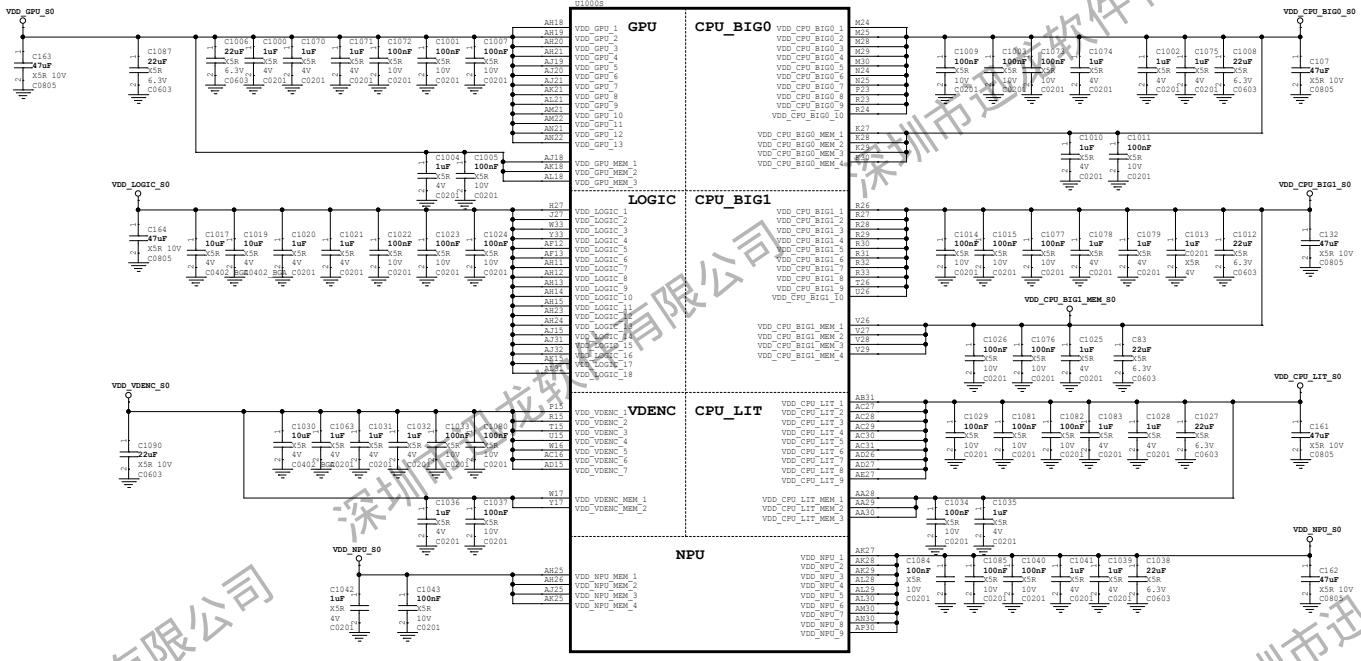
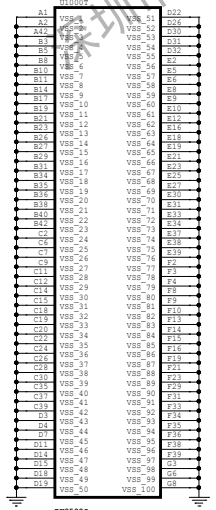
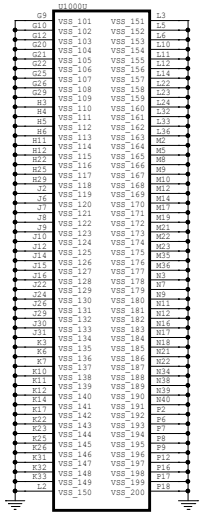
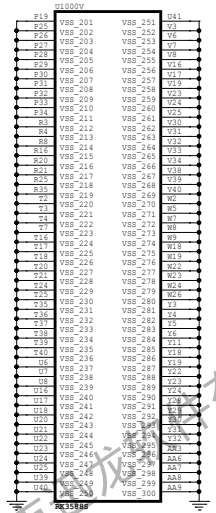
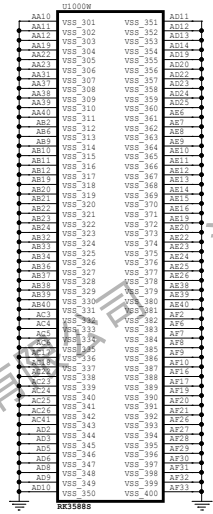
RK3588S(Power&Gnd)

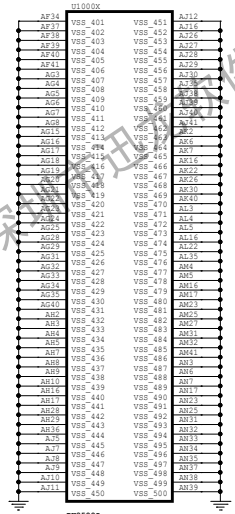


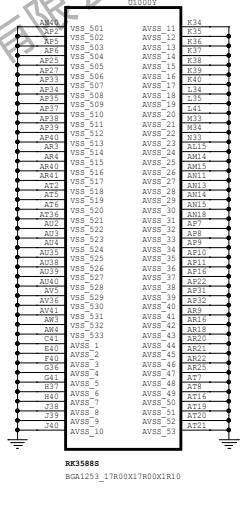


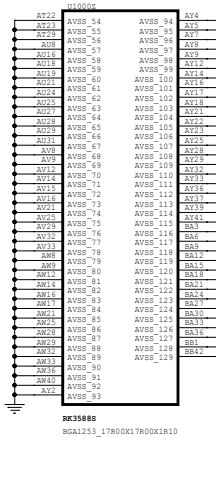








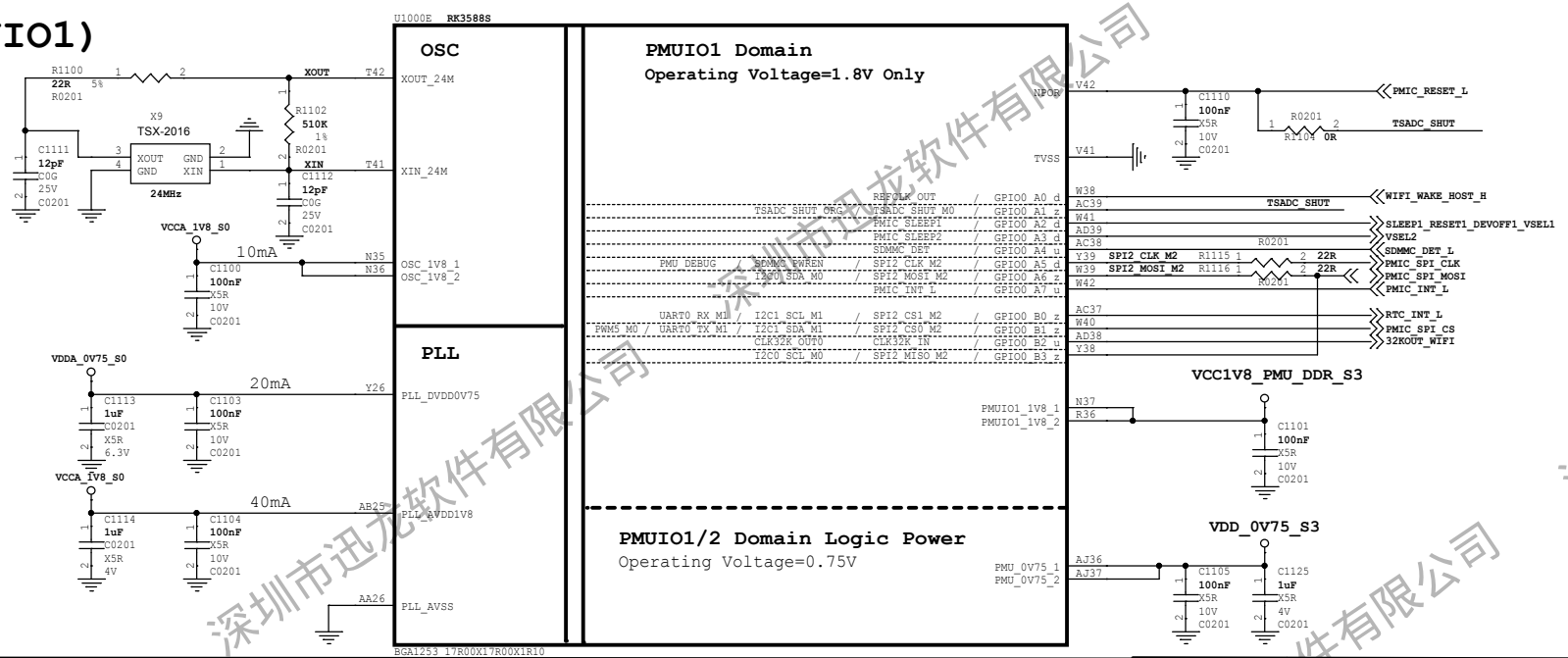




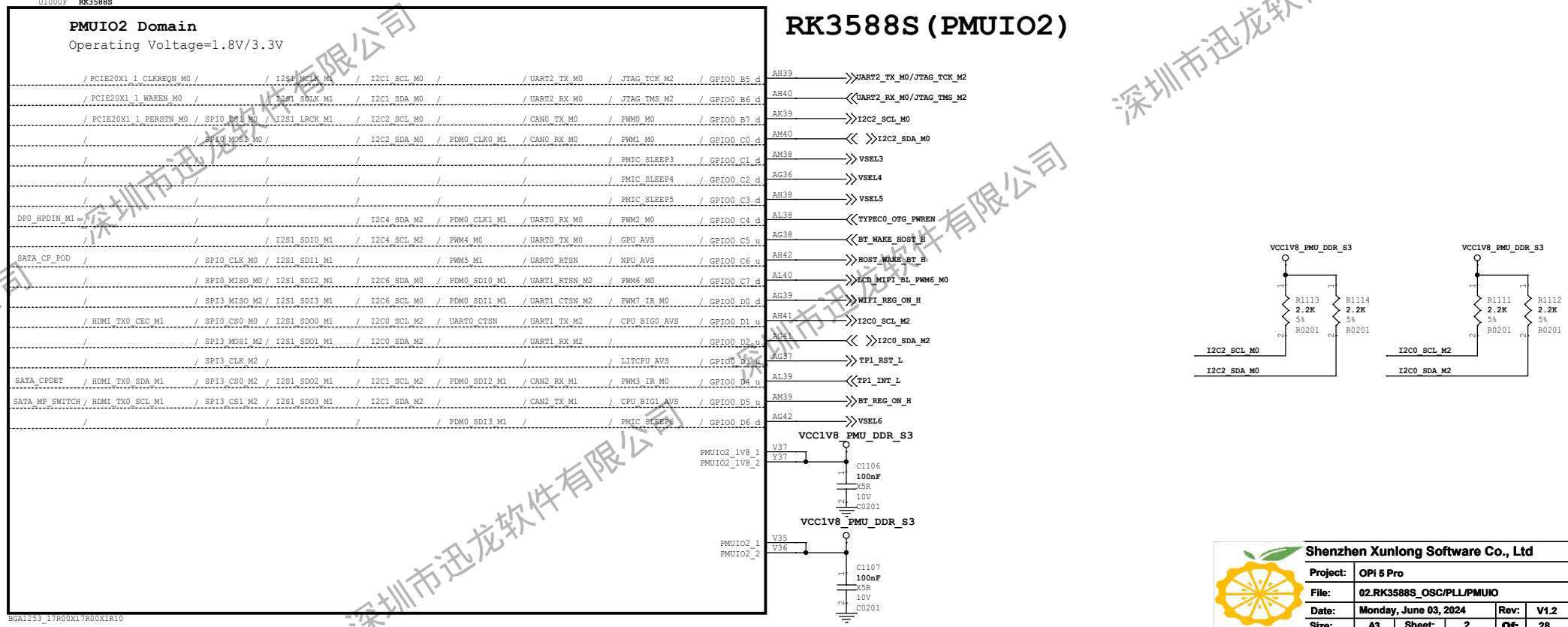
深圳市迅龙软件有限公司

Shenzhen Xunlong Software Co., Ltd					
Project:	OPi 5 Pro				
File:	01.RK3588S_Power/GND				
Date:	Monday, June 03, 2024			Rev:	V1.2
Size:	A3	Sheet:	1	Of:	28

RK3588S(OSC/PLL/PMUIO1)



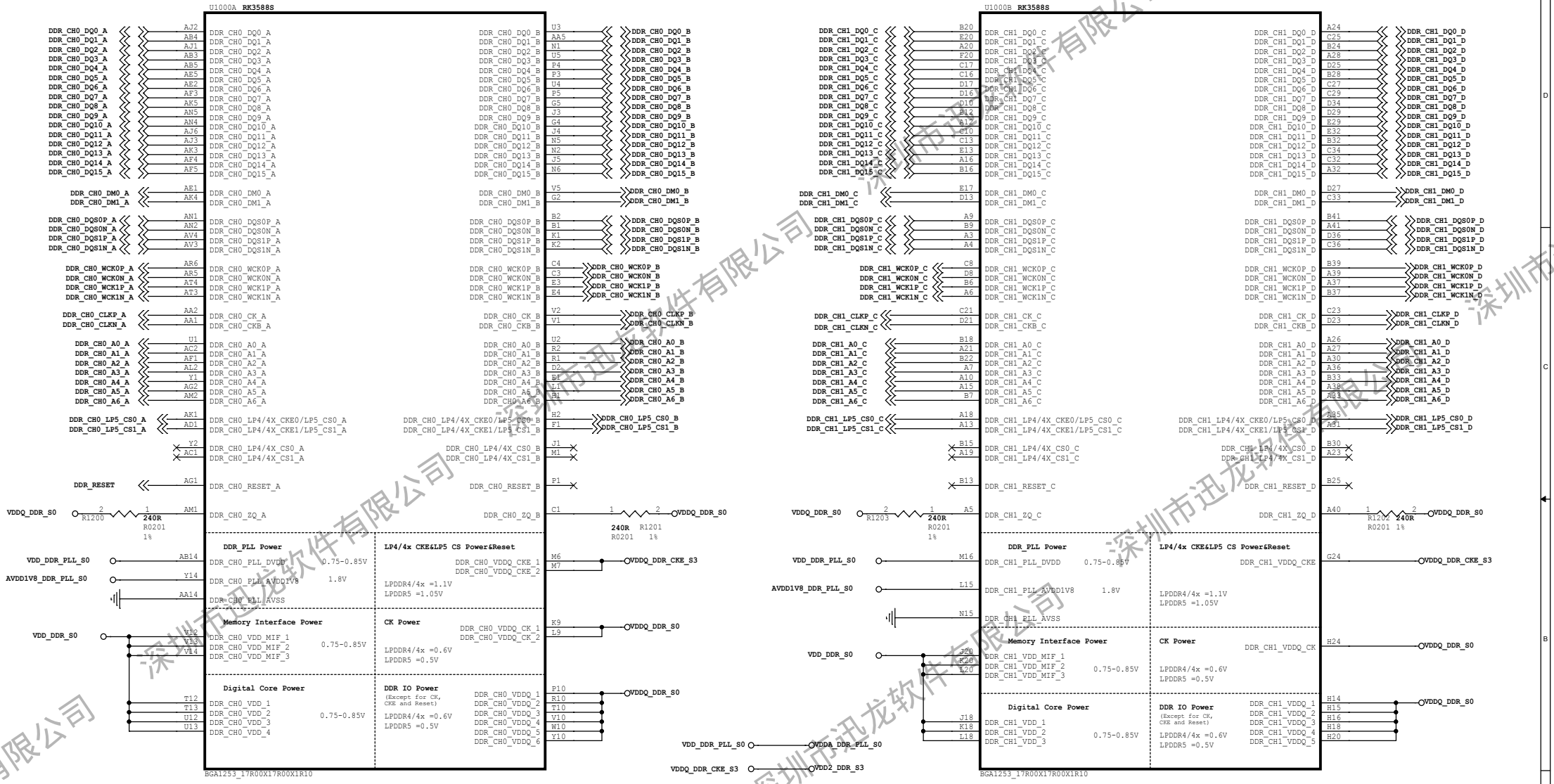
RK3588S(PMUIO2)



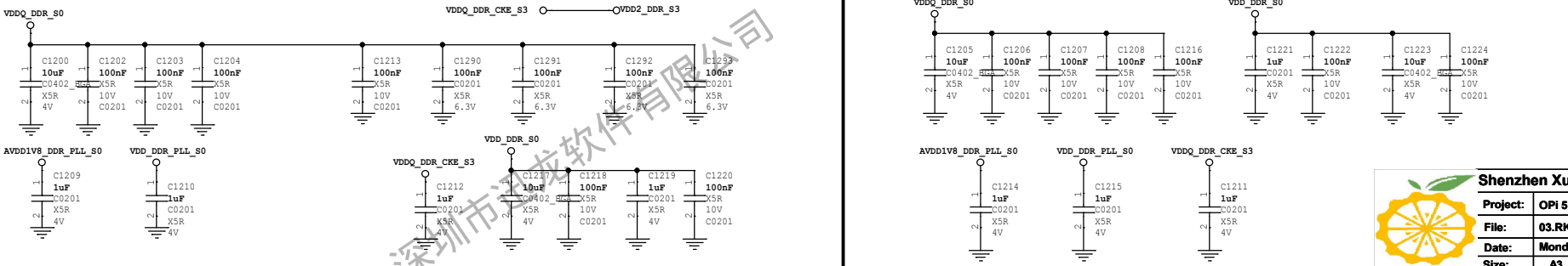
Shenzhen Xunlong Software Co., Ltd					
Project:	OPi 5 Pro				
File:	02.RK3588S_OSC/PLL/PMUIO				
Date:	Monday, June 03, 2024			Rev:	V1.2
Size:	A3	Sheet:	2	Of:	28

深圳市迅龙软件有限公司

RK3588S(DDR PHY)

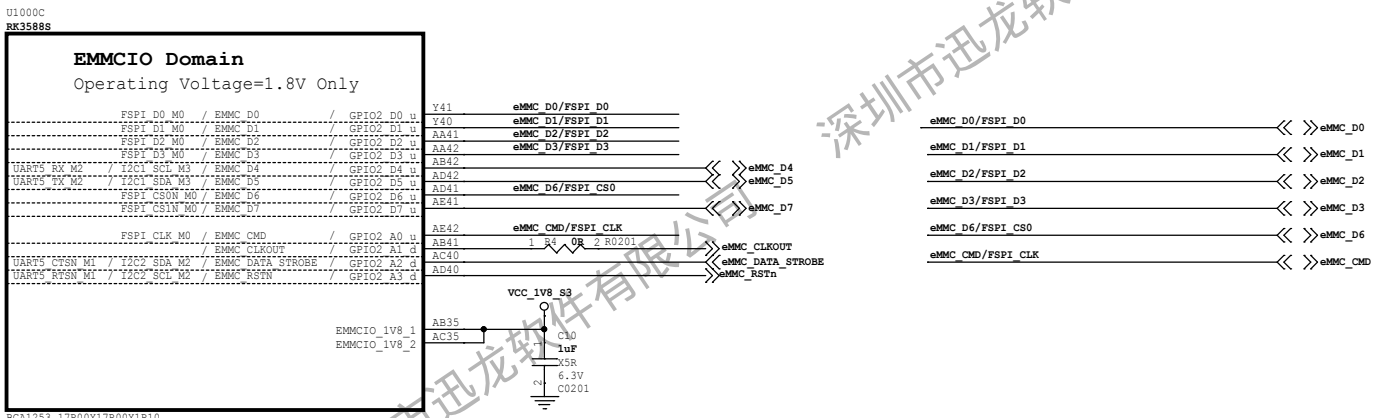


DDR FILTER

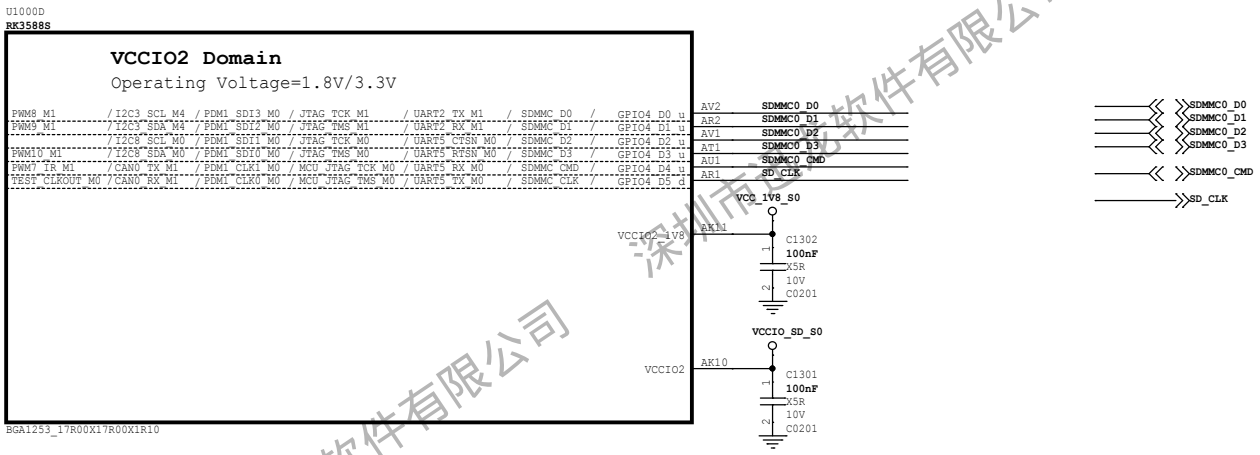


Shenzhen Xunlong Software Co., Ltd					
Project:	OPi 5 Pro				
File:	03.RK3588S DDR Controler				
Date:	Monday, June 03, 2024			Rev:	V1.2
Size:	A3	Sheet:	3	Of:	28

RK3588S(EMMCIO Domain)



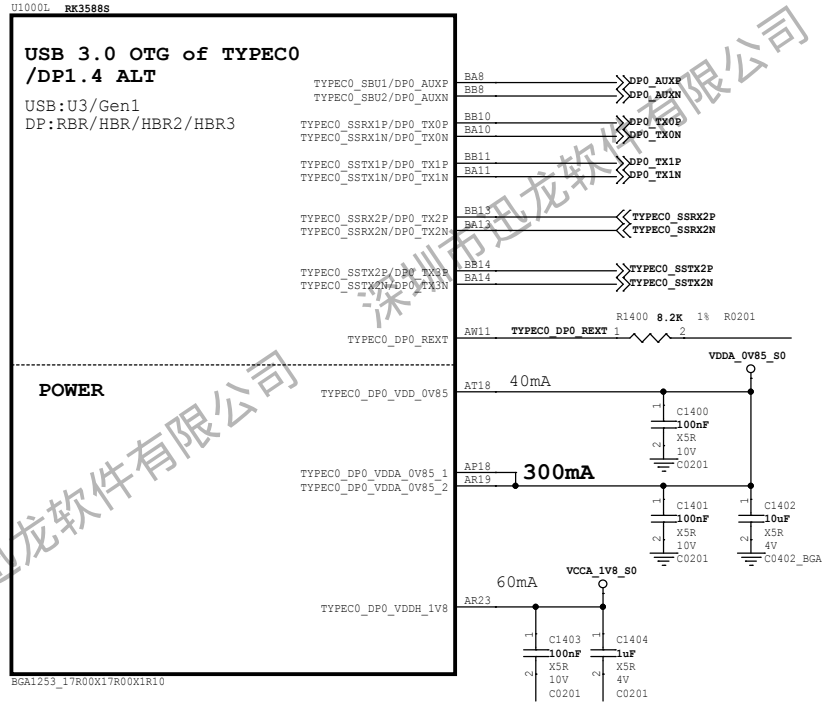
RK3588S(VCCIO2 Domain)



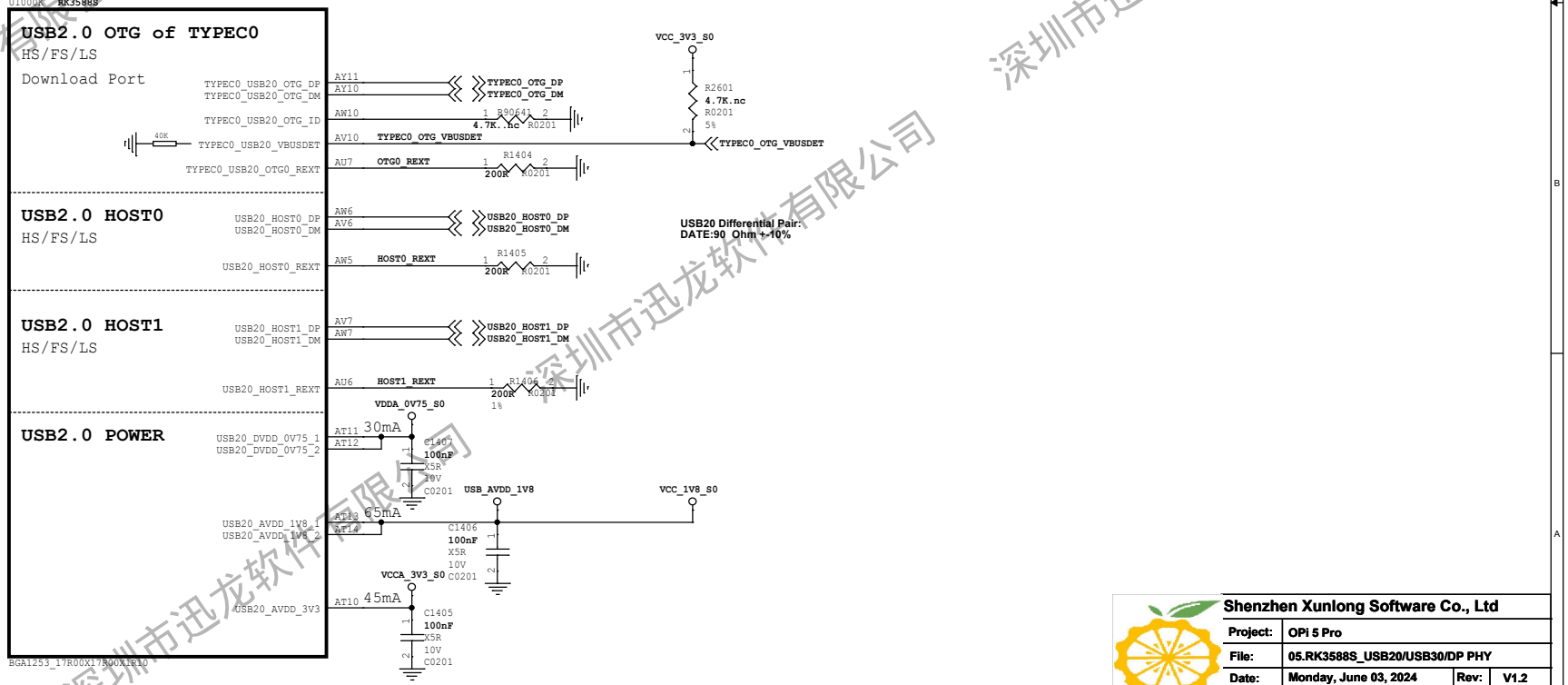
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Shenzhen Xunlong Software Co., Ltd					
Project:	OPi 5 Pro				
File:	04.RK3588S Flash/SD Controller				
Date:	Monday, June 03, 2024			Rev:	V1.2
Size:	A3	Sheet:	4	Of:	28

RK3588S(USB3.0/DP1.4)



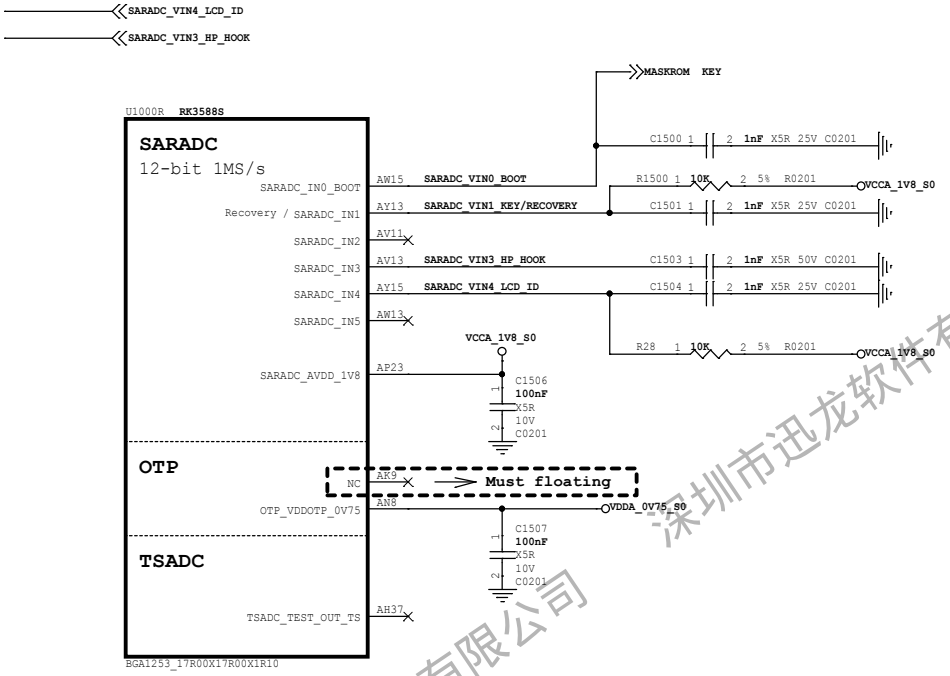
RK3588S(USB2.0)

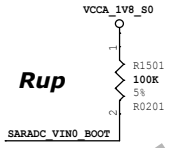


深圳市迅龙软件有限公司

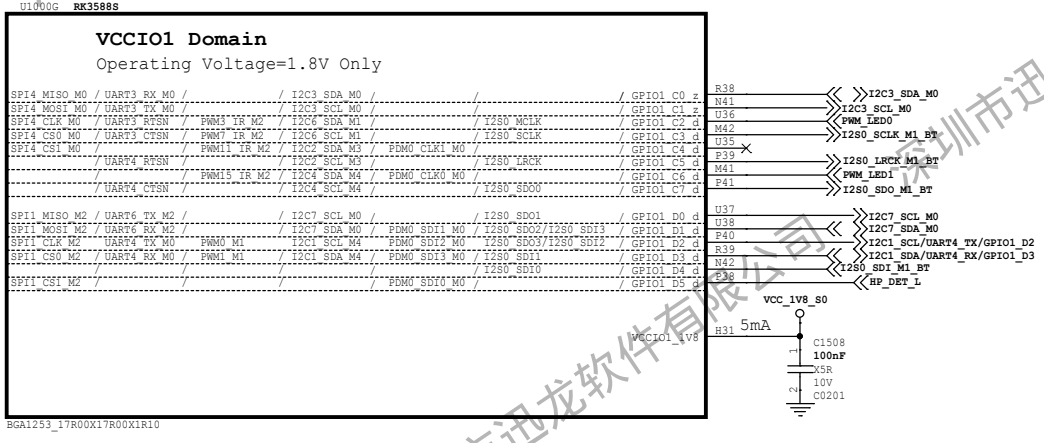
Shenzhen Xunlong Software Co., Ltd					
Project:	OPi 5 Pro				
File:	05.RK3588S_USB20/USB30/DP PHY				
Date:	Monday, June 03, 2024			Rev:	V1.2
Size:	A3	Sheet:	5	Of:	28

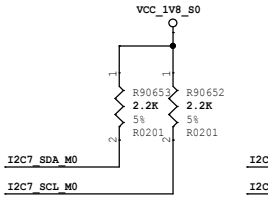
RK3588S(SARADC/OTP/TSADC)

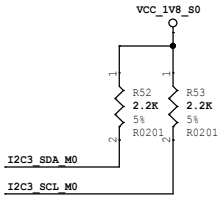




RK3588S(VCCIO1 Domain)



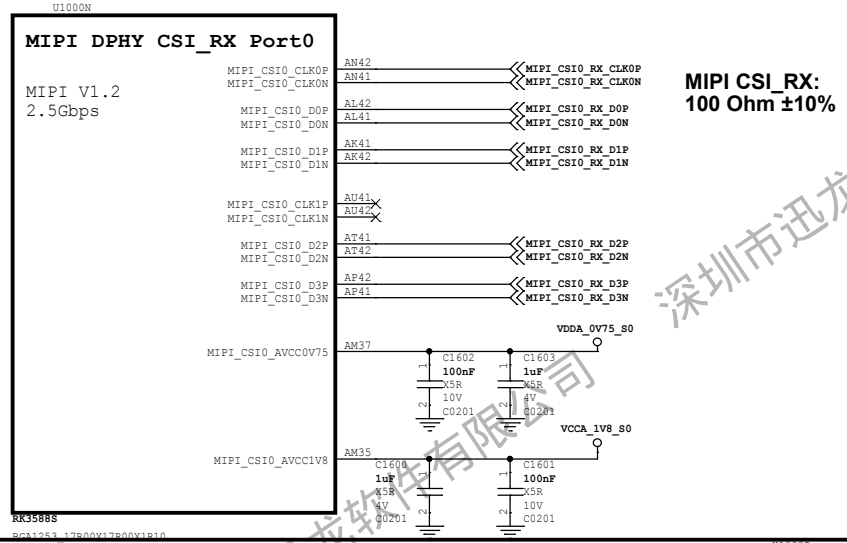




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Shenzhen Xunlong Software Co., Ltd					
Project:	OPi 5 Pro				
File:	06.RK3588S_SARADC/1.8V GPIO				
Date:	Monday, June 03, 2024			Rev:	V1.2
Size:	A3	Sheet:	6	Of:	28

RK3588S(MIPI_CSI)

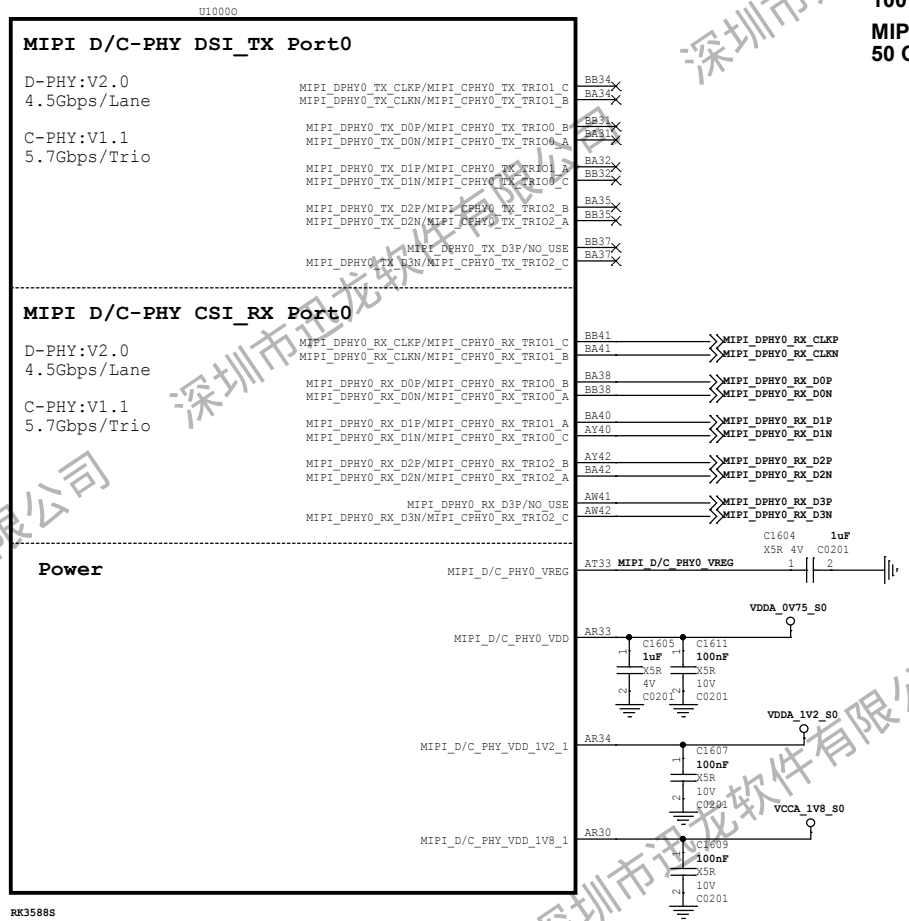


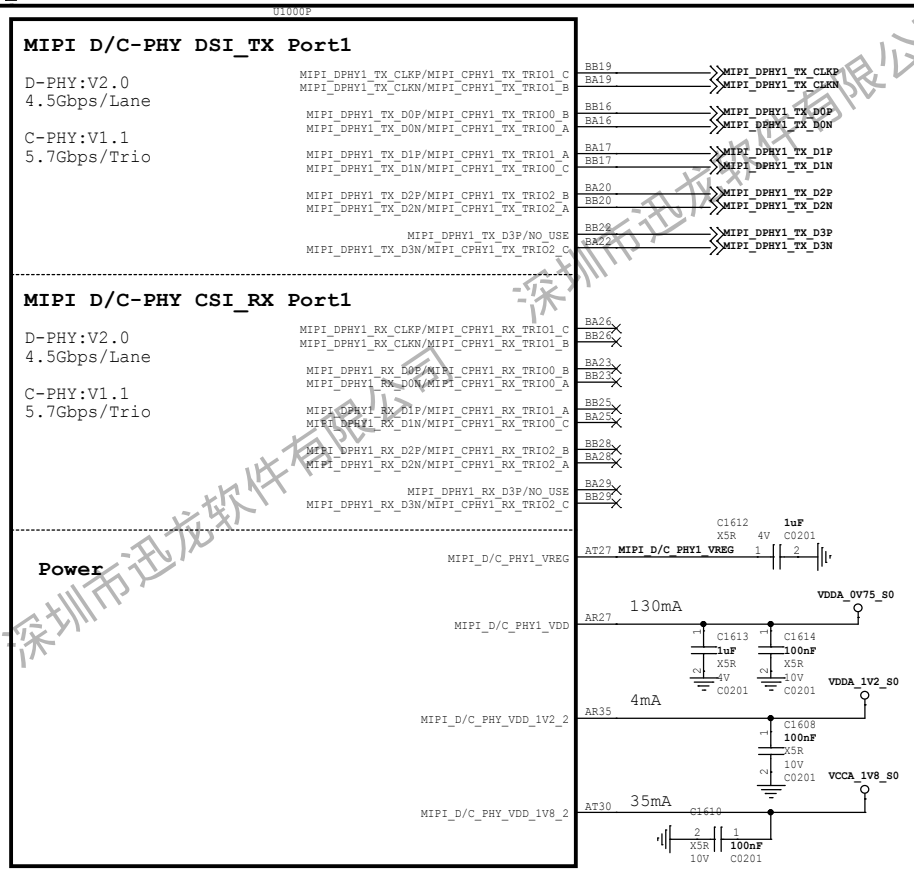
MIPI CSI_RX:
100 Ohm ±10%

RK3588S(MIPI_D/C PHY)

MIPI DPHY:
100 Ohm ±10%

MIPI CPHY:
50 Ohm ±10%

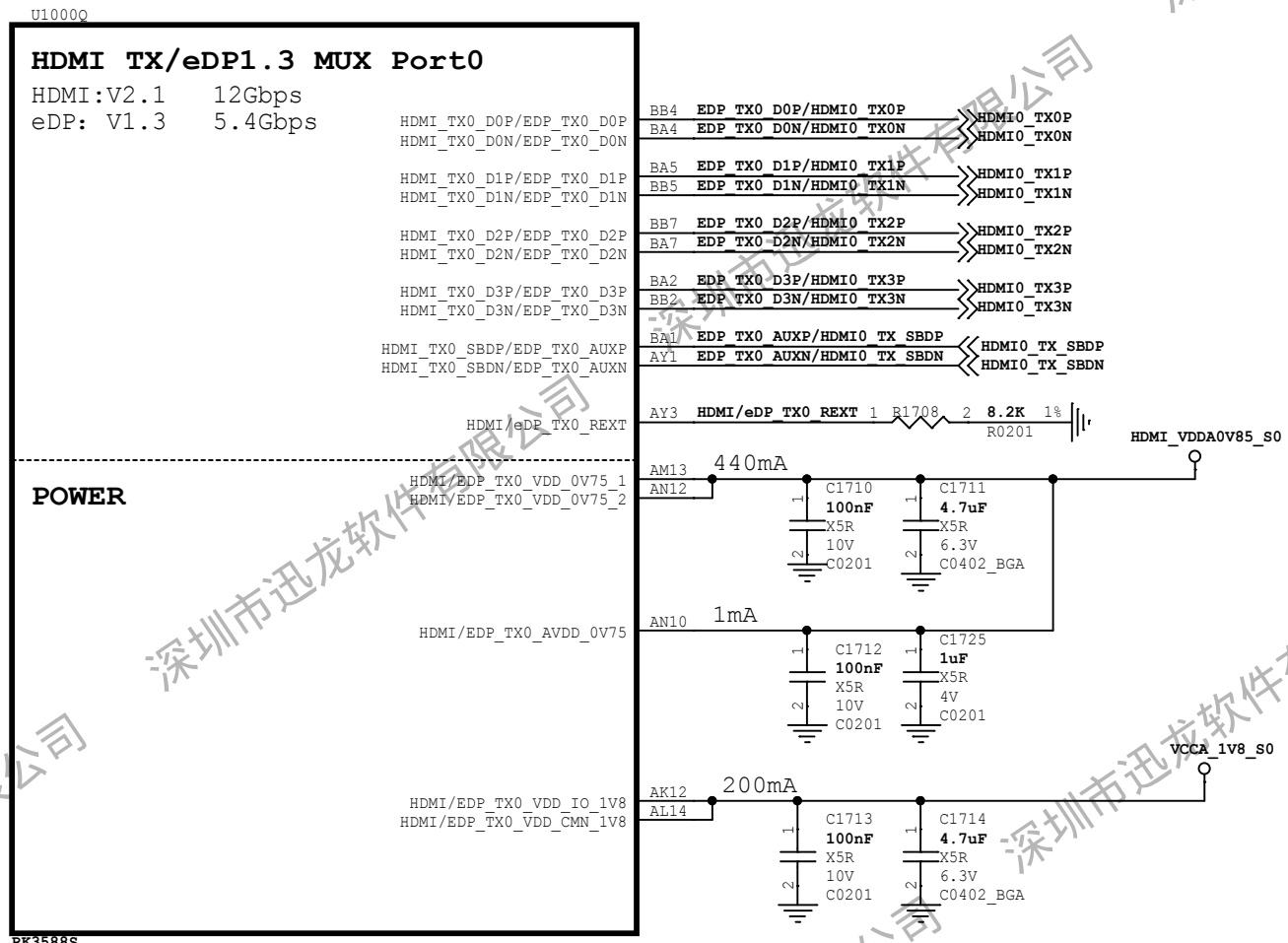




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Shenzhen Xunlong Software Co., Ltd					
Project:	OPi 5 Pro				
File:	07.RK3588S_MIPI Interface				
Date:	Monday, June 03, 2024			Rev:	V1.2
Size:	A3	Sheet:	7	Of:	28

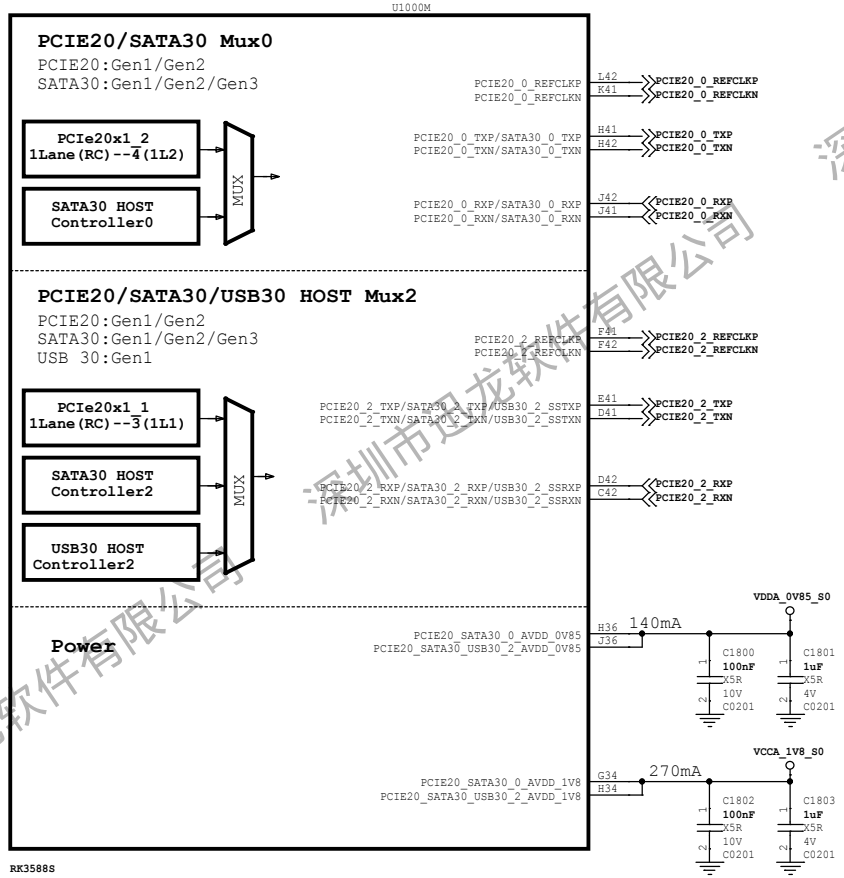
RK3588S(HDMI2.1 TX/eDP1.3 TX)



深圳市迅龙软件有限公司

Shenzhen Xunlong Software Co., Ltd					
Project:	OPi 5 Pro				
File:	08.RK3588S_HDMI/eDP Interface				
Date:	Monday, June 03, 2024			Rev:	V1.2
Size:	A3	Sheet:	8	Of:	28

RK3588S(PCIE20/SATA30/USB30)

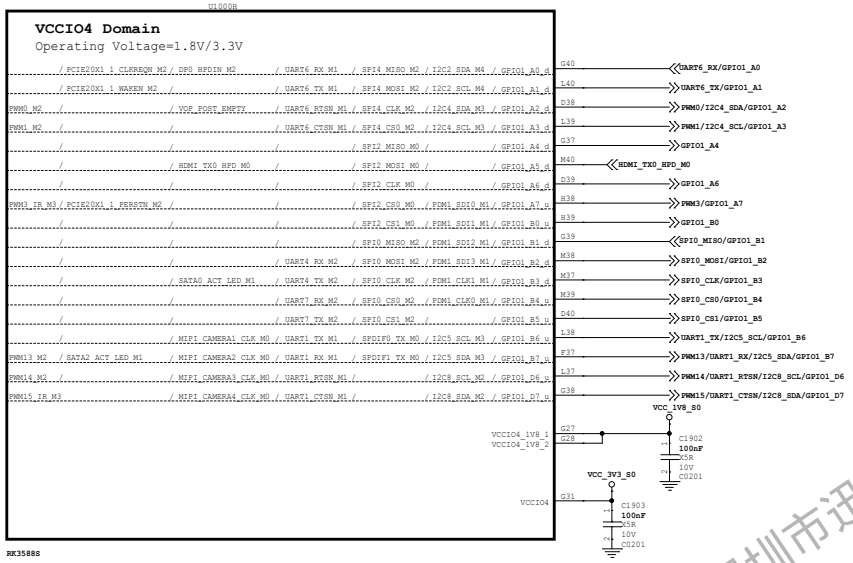


PCIe2.0 PHY

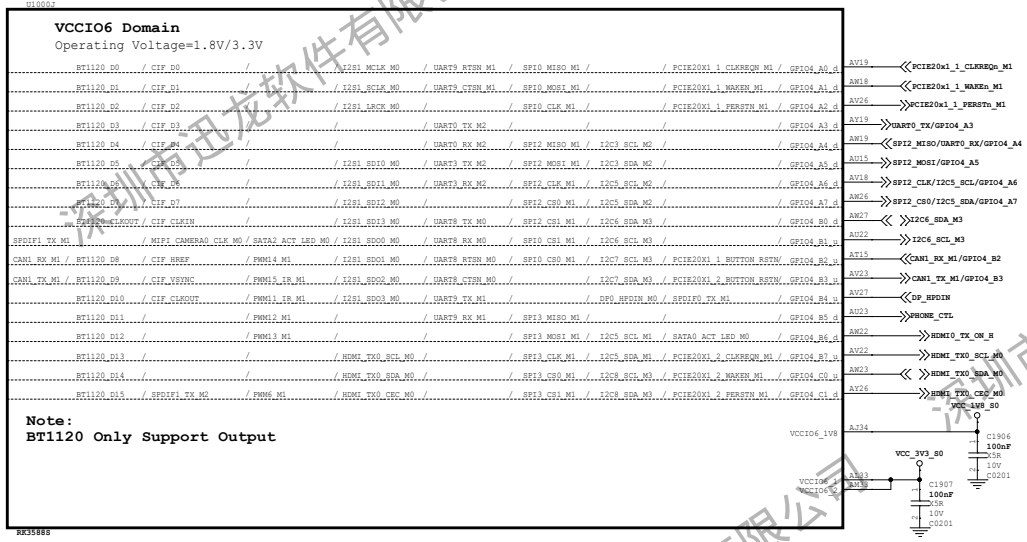
深圳市迅龙软件有限公司

Shenzhen Xunlong Software Co., Ltd					
Project:	OPi 5 Pro				
File:	09.RK3588S PCIE2/SATA3/USB3 PHY				
Date:	Monday, June 03, 2024			Rev:	V1.2
Size:	A3	Sheet:	9	Of:	28

RK3588S(VCCIO4 Domain)



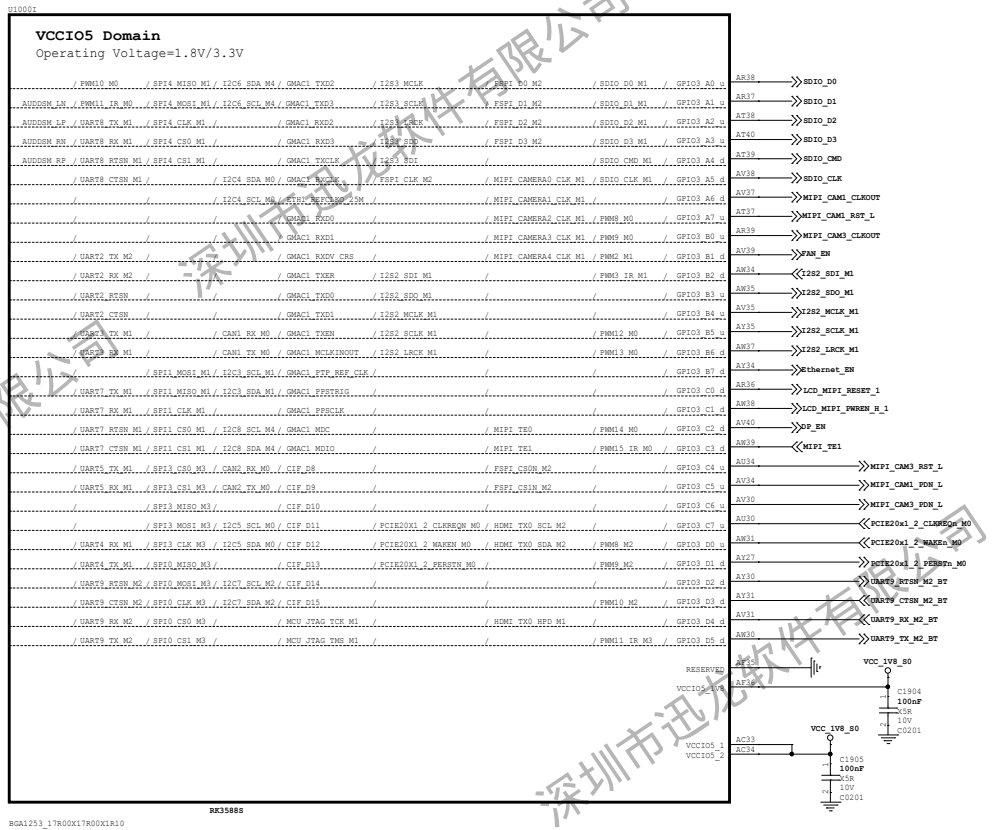
RK3588S(VCCIO6 Domain)



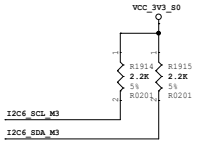
Note:
BT1120 Only Support Output

RK3588S
BGA1253_17R00X17R00X1R10

RK3588S(VCCIO5 Domain)



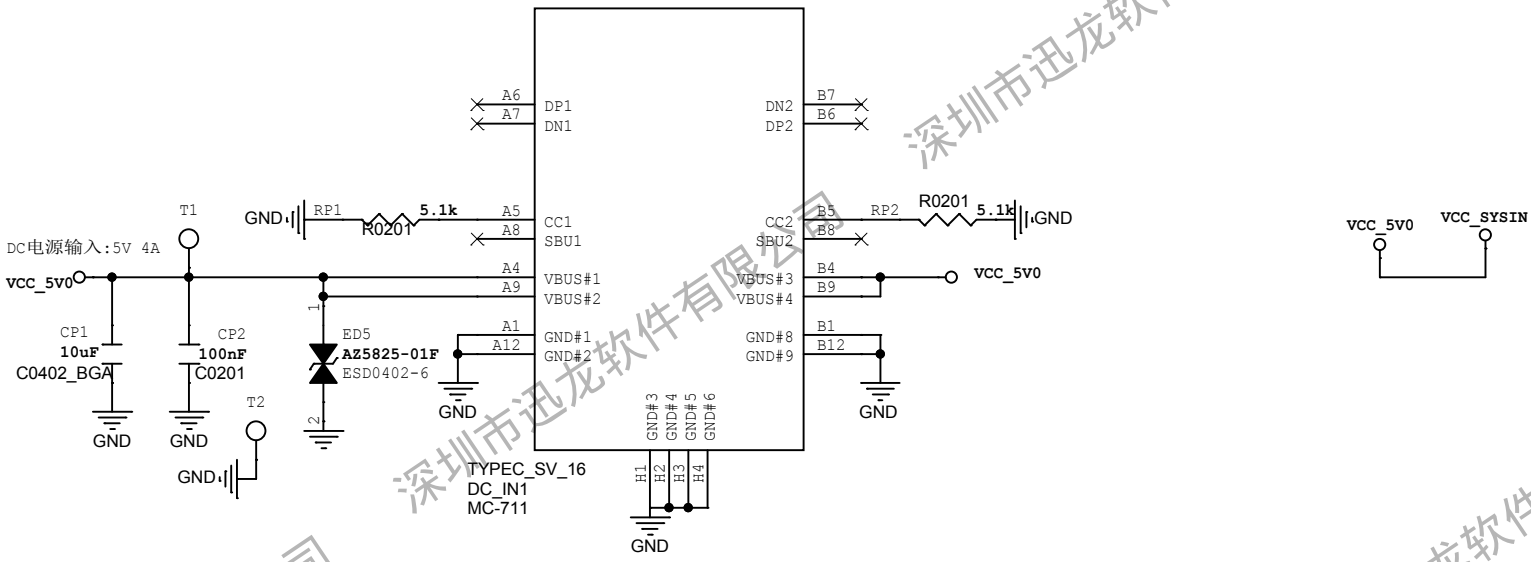
RK3588S
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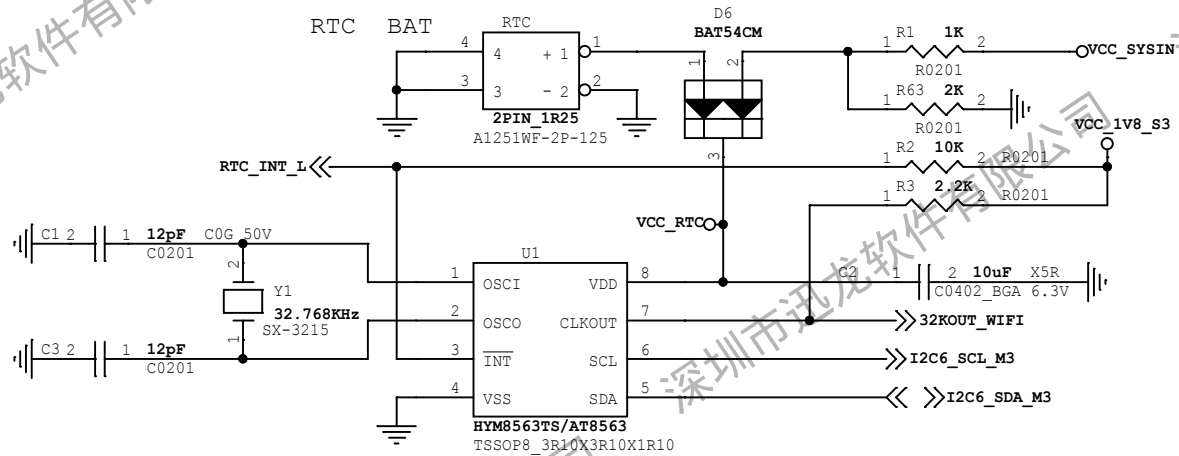


Shenzhen Xunlong Software Co., Ltd					
Project:	OPi 5 Pro				
File:	10.RK3588S GPIO				
Date:	Monday, June 03, 2024		Rev:	V1.2	
Size:	A3	Sheet: 10	Of:	28	

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TPYE-C POWER IN





Address:Read A3H,Write A2H

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Shenzhen Xunlong Software Co., Ltd					
Project:	OPi 5 Pro				
File:	11.Power_Type-C RTC				
Date:	Monday, June 03, 2024			Rev:	V1.2
Size:	A3	Sheet:	11	Of:	28

PMIC1 RK806-1

SLEEP1_RESET1_DEVOFF1_VSEL1
VSEL4
VSEL5
PMIC_INT_L
PMIC_RESET_L
PMIC_EXT_EN_OUT
PMIC_SPI_CLK
PMIC_SPI_CS
PMIC_SPI_MOSI

VCC_SYSIN
ED2200
AZ5825-01F
ESD0402-6

IF TVS UNMOUNTED, ESD OR SURGE SHOULD BE DAMAGE THE PMIC!!!

This device must be mounted.Replacing TVS mode is not recommended, if must, please choose the same specifications
Operating Supply Vltage : 5.5V(5.25-6V)
PeakPulse Current: >10A (tp=8/20uS)
Surge Clamping Voltage: <6.5V

DO NOT DELETE IT!

VSEL5 — PMIC_PWR_CTRL2
VSEL4 — PMIC_PWR_CTRL3

PMIC RK806-1 BUCK

U2200A

VDD_DDR_S0 — C2200 22uF/10V X5R C0603 — L2200 0.47uH IND_252012 — PMIC_SW5 — BUCK5 2.5A — VOUT5 — FB5 (FB=0.5V)
R2200 100R 5% R0201; C9922 1uF X5R 6.3V C0201
VDD_DDR_S0 Feedback from RK3588S

VDD2_DDR_S3 — T11 — C2214 22uF X5R 10V C0603; C2215 22uF X5R 10V C0603; C2216 100pF NPO 25V C0201 — L2202 0.47uH IND_252012 — PMIC_SW6 — BUCK6 2.5A — VOUT6 — PMIC_VOUT6 — PMIC_FB6 — FB6 (FB=0.5V)
R2218 110K 1% R0201; R2227 100K 1% R0201

LPDDR4/4x=1.1V	120K
LPDDR5=1.05V	110K

VDDQ_DDR_S0 — T60 — C2228 22uF X5R 10V C0603; C2227 22uF X5R 10V C0603; C2229 100pF NPO 25V C0201 — L2204 0.47uH IND_201610 — PMIC_SW9 — BUCK9 2.5A — VOUT9 — PMIC_VOUT9 — PMIC_FB9 — FB9 (FB=0.5V)
R2239 0R R0201; R2241 NC 1% R0201

LPDDR4/4x=0.6V	20K	100K
LPDDR5=0.5V	0R	DNP

VCC_3V3_S3 — T33 — C2238 22uF X5R 10V C0603; C2239 22uF X5R 10V C0603 — L2206 0.47uH IND_252012 — PMIC_SW8 — BUCK8 2.5A — VOUT8 — PMIC_VOUT8

VDD1_1V8_DDR_S3 — VCC1V8_PMU_DDR_S3 — VCC_1V8_S3 — T18 — C2242 22uF X5R 10V C0603; C2243 22uF X5R 10V C0603 — L2208 0.47uH IND_201610 — PMIC_SW10 — BUCK10 2.5A — VOUT10 — PMIC_VOUT10
Default:1.8V

RK806-1 QFN68_7R00X7R00X0R80_T

BUCK1 6.5A — VCC_SYSIN — C2201 22uF/10V X5R C0603 — L2201 0.22uH IND_404020 — PMIC_SW1 — PMIC_VOUT1 — VDD_GPU_S0 — T85 — C2209 1uF X5R 6.3V C0201; R2208 100R 5% R0201; C2210 22uF X5R 10V C0603; C2205 47uF X5R 10V C0805; C2280 47uF X5R 10V C0805; C94 100nF C0201; C95 100nF C0201; C96 100nF C0201
VDD_GPU_S0 Feedback from RK3588S

BUCK2 5A — VCC_SYSIN — C2213 22uF/10V X5R C0603 — L2203 0.24uH IND_252012 — PMIC_SW2 — PMIC_VOUT2 — VDD_CPU_LIT_S0 — T95 — C2221 1uF X5R 6.3V C0201; R2219 100R 5% R0201; C68 47uF X5R 10V C0805; C69 47uF X5R 10V C0805; C105 100nF C0201; C106 100nF C0201
VDD_CPU_LIT_S0 Feedback from RK3588S

BUCK3 5A — VCC_SYSIN — C2226 22uF/10V X5R C0603 — L2205 0.24uH IND_252012 — PMIC_SW3 — VDD_LOGIC_S0 — T75 — C2230 1uF X5R 6.3V C0201; R2238 100R 5% R0201; C2231 22uF X5R 10V C0603; C165 47uF X5R 10V C0805; C99 100nF C0201; C100 100nF C0201
VDD_LOGIC_S0 Feedback from RK3588S

BUCK4 5A — VCC_SYSIN — C2236 22uF/10V X5R C0603 — L2207 0.24uH IND_252012 — PMIC_SW4 — VDD_VDENC_S0 — T77 — C2291 1uF X5R 6.3V C0201; R2209 100R 5% R0201; C2240 22uF X5R 10V C0603; C2237 22uF X5R 10V C0603; C2241 22uF X5R 10V C0603; C102 100nF C0201; C103 100nF C0201
VDD_VDENC_S0 Feedback from RK3588S

BUCK7 2.5A — VCC_SYSIN — L2209 0.47uH IND_201610 — PMIC_SW7 — PMIC_VOUT7 — T20 — VCC_2V0_PLDO_S3 — C2244 22uF X5R 10V C0603; C2245 22uF X5R 10V C0603

PMIC RK806-1 Managerment

U2200C

VCC_1V8_S3_PLDO6 — C2248 1uF 6.3V X5R C0201

VCCIO
(IIC/SPI)CS — PMIC_SPI_CS
(SDA)MOSI — PMIC_SPI_MOSI
(SCL)CLK — PMIC_SPI_CLK
(PWRCTRL3)SO — PMIC_PWR_CTRL3
PWRCTRL2 — PMIC_PWR_CTRL2
PWRCTRL1 — SLEEP1_RESET1_DEVOFF1_VSEL1
R2258 100K 5% R0201

VCCA
SYNC_CLK — PMIC_EXT_EN_OUT
SYNC — PMIC_INT_L
VDC — PMIC_RESET_L
PWRON — ePAD
(M/S)EXT_EN — INT — RESETB
R2261 10K 5% R0201; R2263 10K 5% R0201

VCC_SYSIN — C2254 22uF/10V X5R C0603
VCC_1V8_S3
C4 10uF C0402_BGA X5R
R2264 100K R0201
PMIC_VDC
R2265 200K C0201; C2260 1nF C0201 X5R 25V
C2261 100nF X5R 10V C0201
C2262 100nF X5R 10V C0201
POWER KEY
RESET — PMIC_RESET_L

RK806-1 QFN68_7R00X7R00X0R80_T

20220929

RK3588S nPOR — TSADC_SHUT
RK806-1 RESETB
Reset Key

Note:
- Reset Key Control Path
- TSADC_SHUT Control Path
- RK806-1 Control Path

PMIC RK806-1 LDO

U2200B

VCC_2V0_PLDO_S3 — C2247 1uF X5R 6.3V C0201
VCC_SYSIN — C2252 22uF/10V X5R 10V C0603
VCC_1V1_NLDO_S3 — C2257 1uF X5R 6.3V C0201
VCC_1V1_NLDO_S3 — C2264 1uF X5R 6.3V C0201
VDD2_DDR_S3 — R5 0R NC R0402

PLDO
VCC11 — 0.5A — PLDO1 — VCCA_1V8_S0 — C2246 10uF 6.3VC0402 X5R
0.3A — PLDO2 — VCC_1V8_S0 — C2249 10uF 6.3VC0402 X5R
0.3A — PLDO3 — VDDA_1V2_S0 — C2250 1uF 6.3VC0201 X5R
VCC12 — 0.5A — PLDO4 — VCCA_3V3_S0 — C2251 10uF 6.3VC0402 X5R
0.3A — PLDO5 — VCCIO_SD_S0 — C2253 1uF 6.3VC0201 X5R

NLDO
VCC13 — 0.3A — NLDO1 — VDD_0V75_S3 — C2255 10uF 6.3VC0402 X5R
0.3A — NLDO2 — VDDA_DDR_PLL_S0 — C2258 10uF 6.3VC0402 X5R
VCC14 — 0.5A — NLDO3 — HDMI_VDDA0V85_S0 — C2259 10uF 6.3VC0402 X5R
0.5A — NLDO4 — VDDA_0V85_S0 — C2263 10uF 6.3VC0402 X5R
0.3A — NLDO5 — VDDA_0V75_S0 — C108 10uF 6.3VC0402 X5R

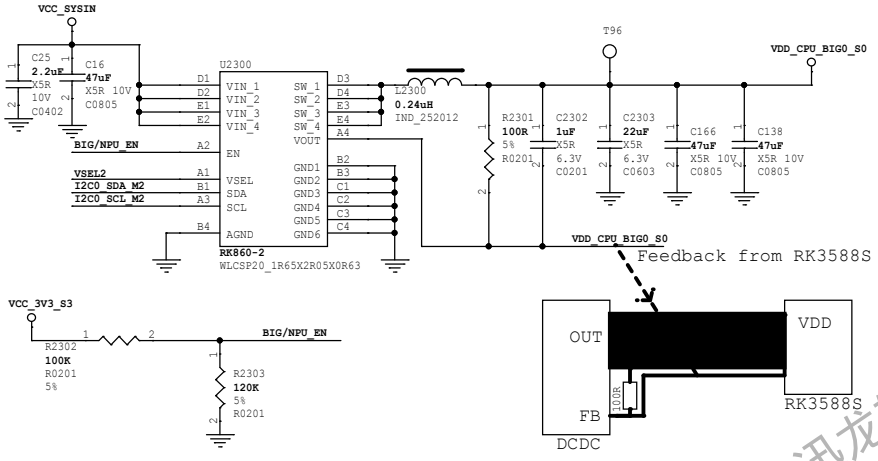
RK806-1 QFN68_7R00X7R00X0R80_T

VCCA_1V8_S0 — AVDD1V8_DDR_PLL_S0 — C2290 1uF X5R 6.3V C0201

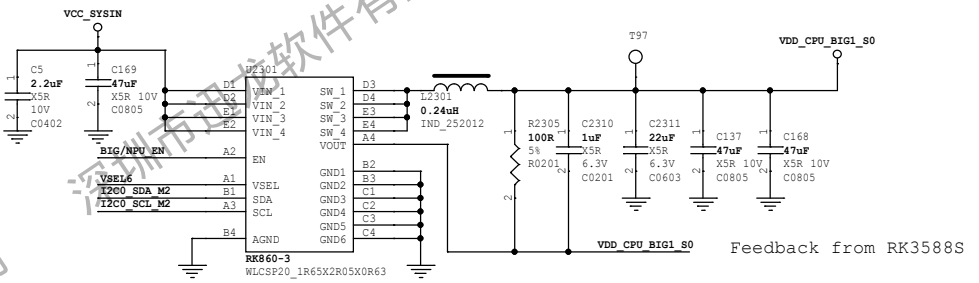
Shenzhen Xunlong Software Co., Ltd				
Project:	OPi 5 Pro			
File:	12.Power-PMIC_RK806-1			
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Size:	A3	Sheet: 12	Of:	28

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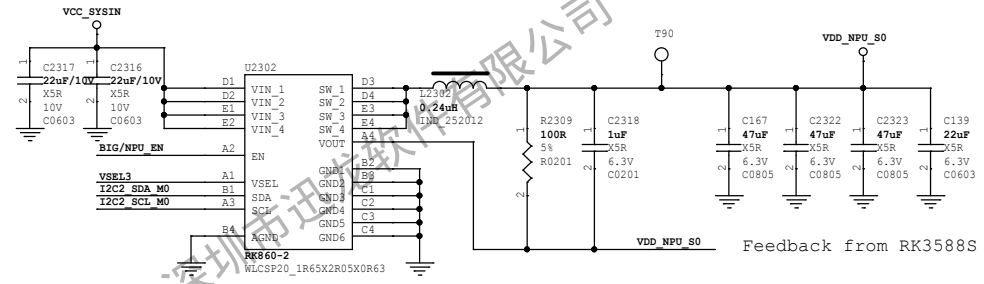
VDD_CPU_BIG0



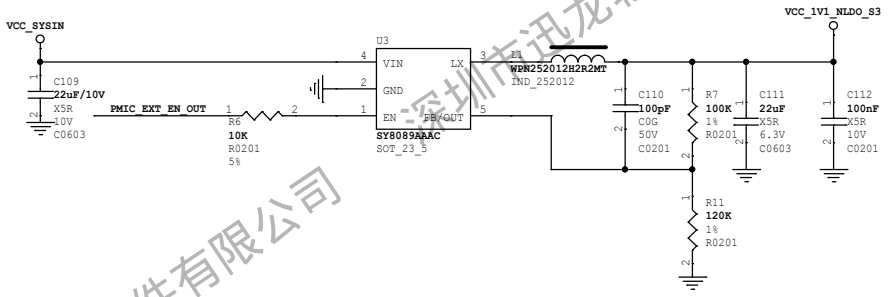
VDD_CPU_BIG1



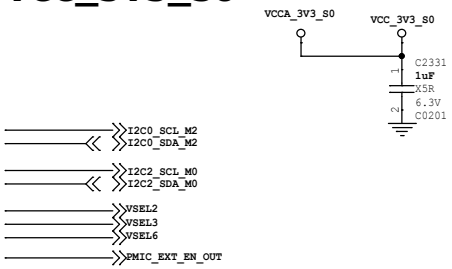
VDD_NPU

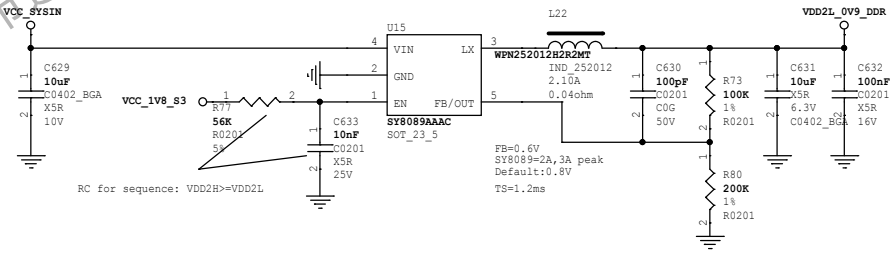


VCC_1V1_NLDO



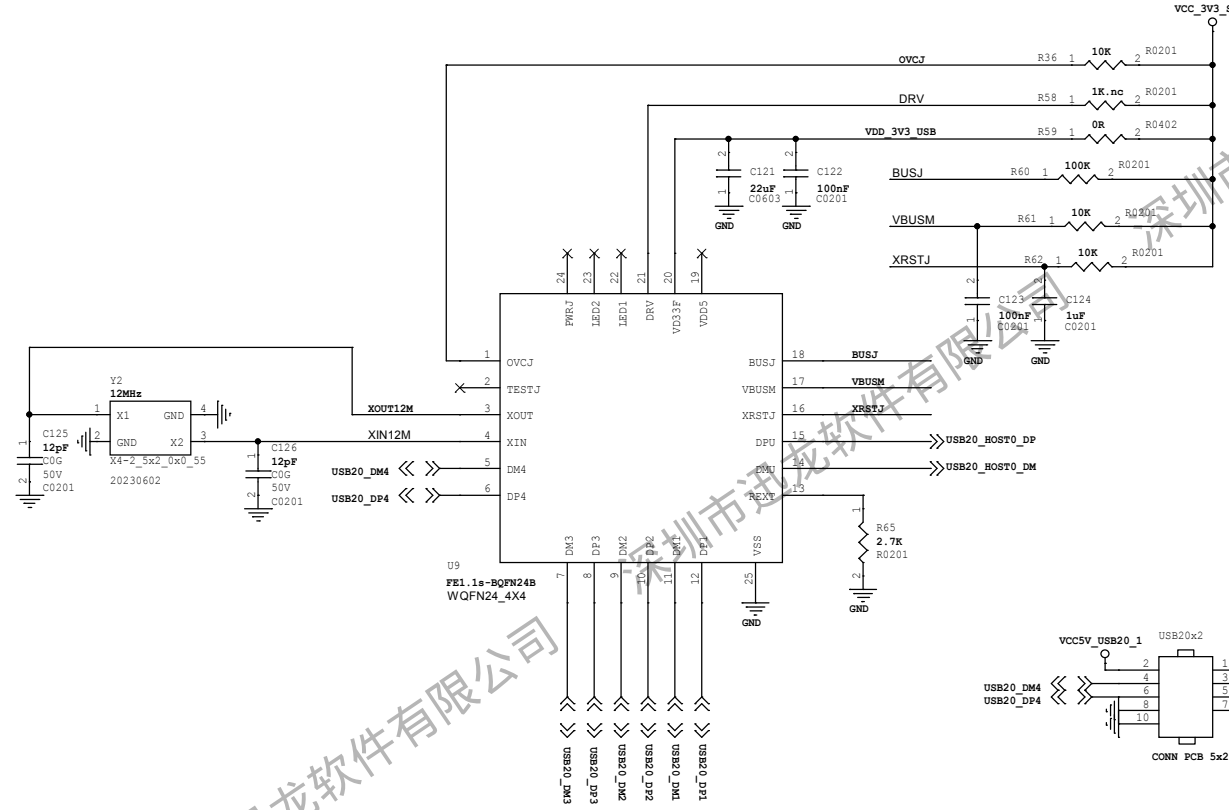
VCC_3V3_S0

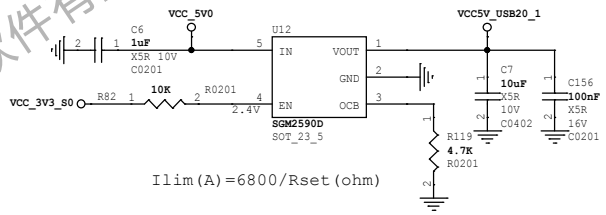


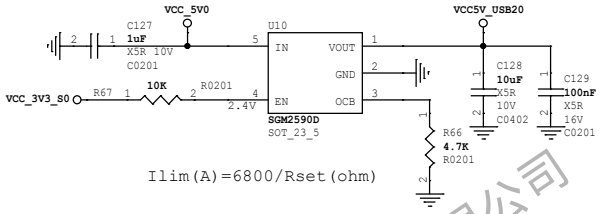


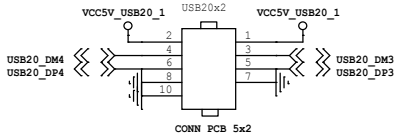
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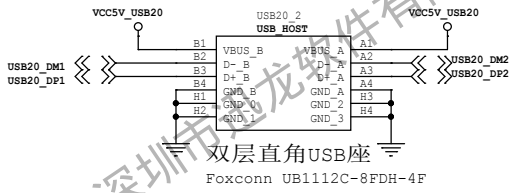
Shenzhen Xunlong Software Co., Ltd					
Project:	OPi 5 Pro				
File:	13.Power_Ext Discrete				
Date:	Monday, June 03, 2024			Rev:	V1.2
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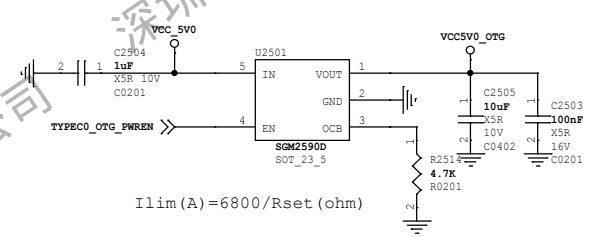


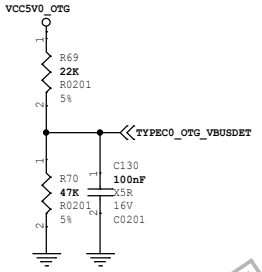


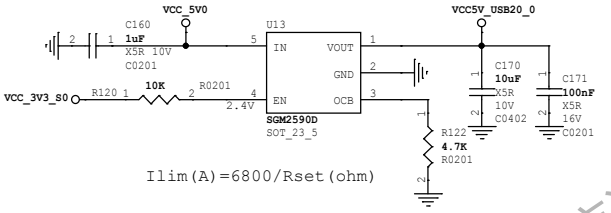


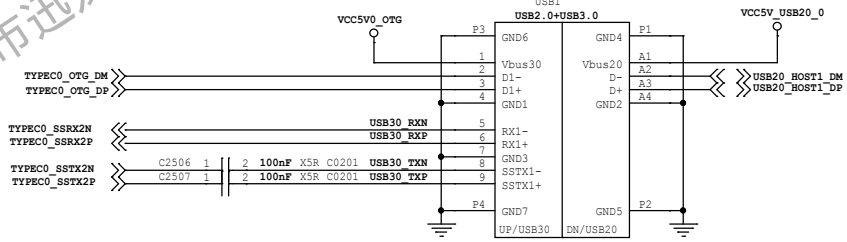
双层直角USB座

Foxconn UB1112C-8FDH-4F









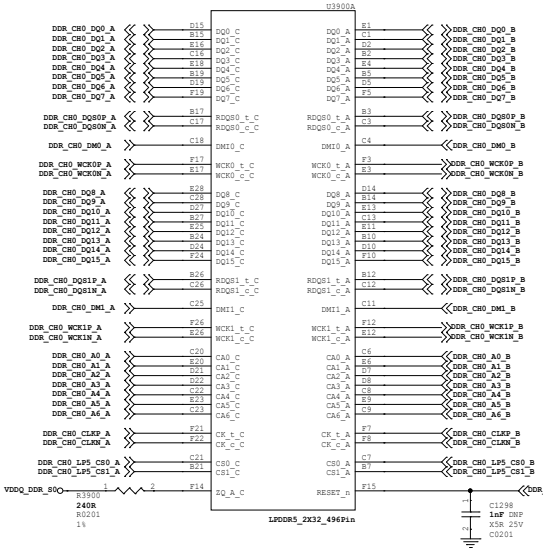
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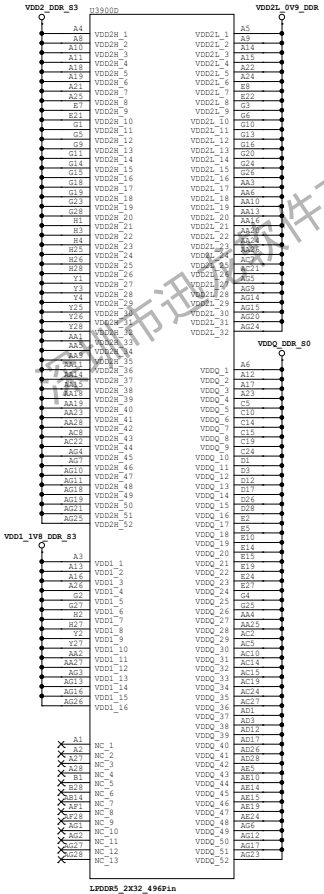
Shenzhen Xunlong Software Co., Ltd					
Project:	OPi 5 Pro				
File:	14.USB20/USB30 HOST Port				
Date:	Monday, June 03, 2024			Rev:	V1.2
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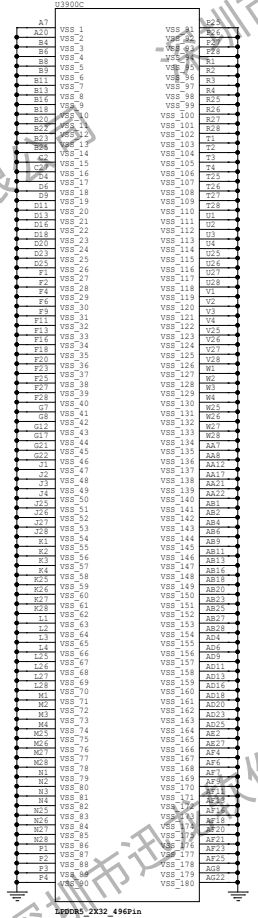
DRAM-LPDDR5X_2X32bit

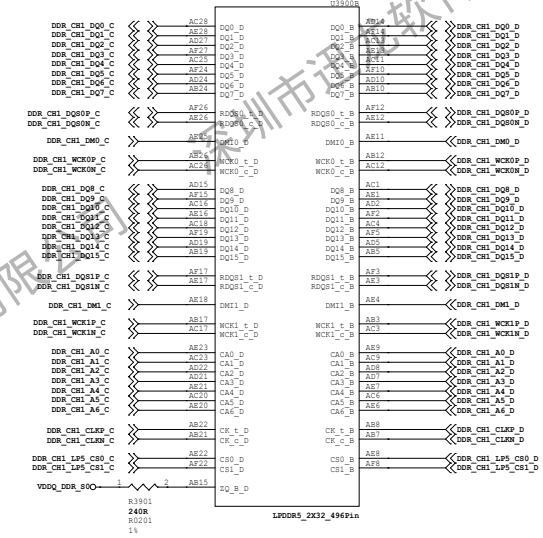
Power sequence: VDD1>VDD2H>=VDD2L>VDDQ-200mV

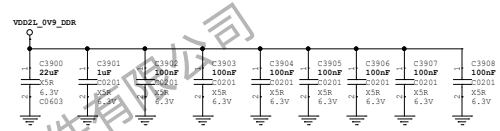
VDD1: 1.70V--1.95V
VDD2H:1.01V--1.12V
VDD2L:0.87V--0.97V
VDDQ: 0.47V--0.57V

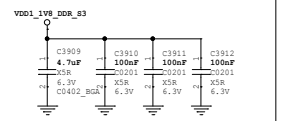


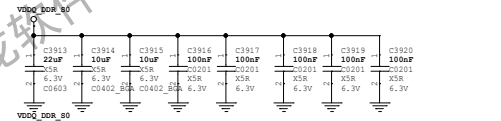


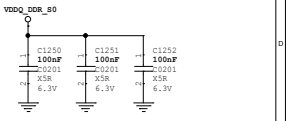


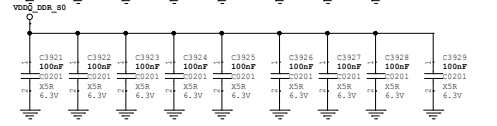


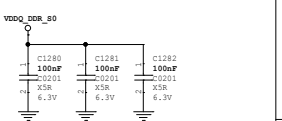


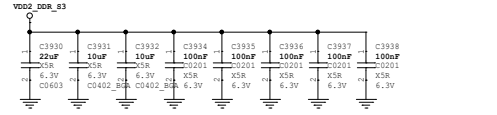


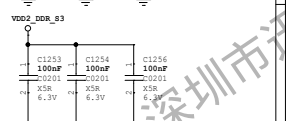


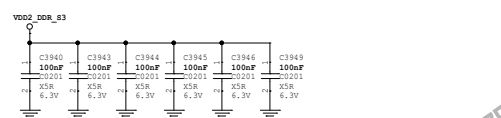


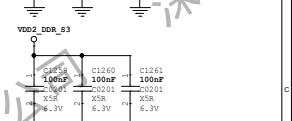


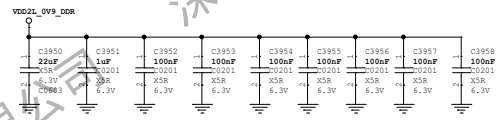


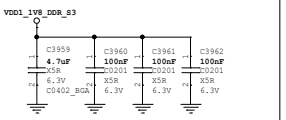


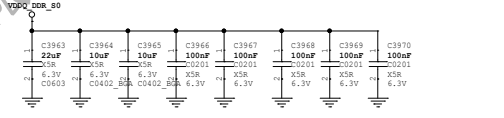


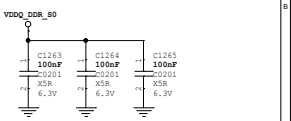


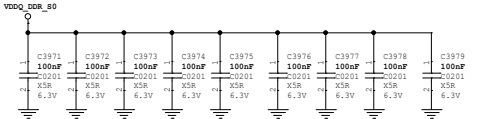


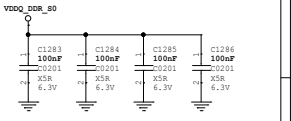


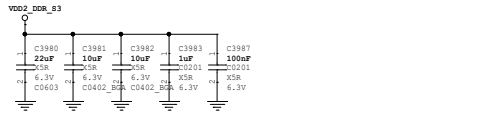


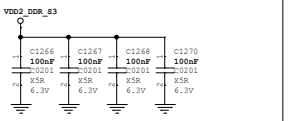


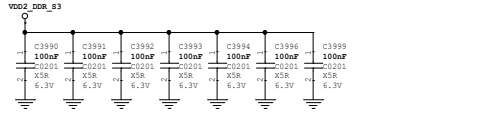


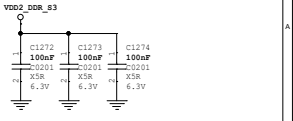








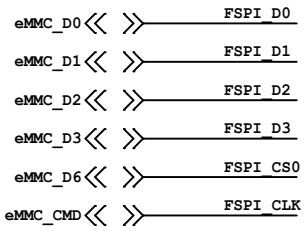


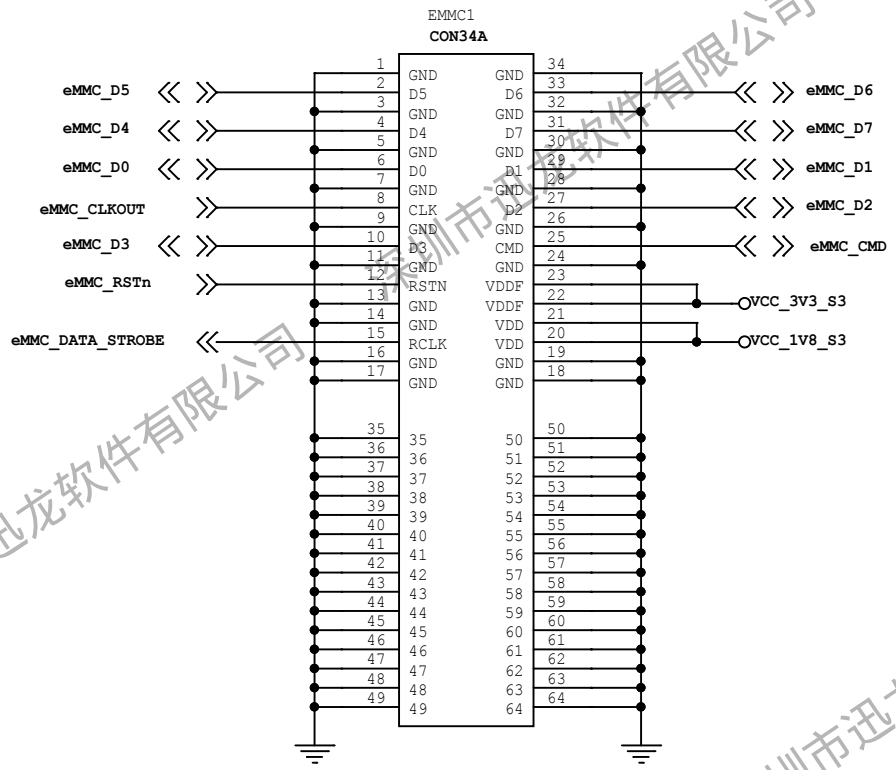


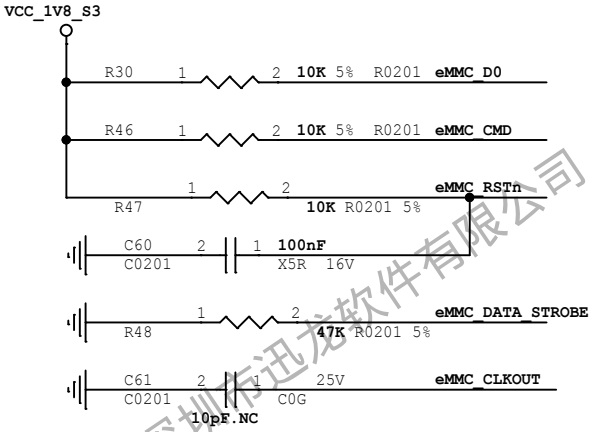
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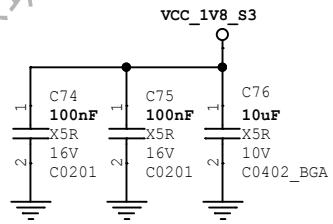
Shenzhen Xunlong Software Co., Ltd				
Project:	OPi 5 Pro			
File:	15.DRAM-LPDDR5X_496P_2X32bit			
Date:	Monday, June 03, 2024		Rev:	V1.2
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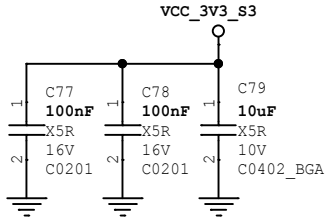
eMMC FLASH

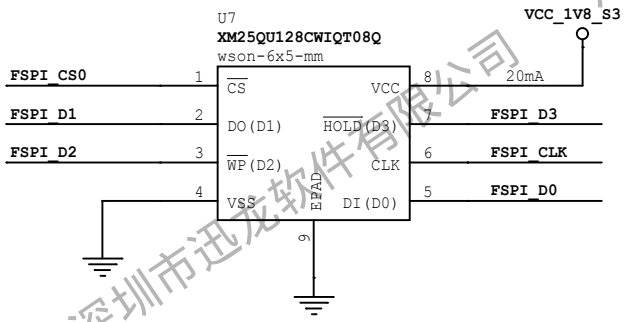












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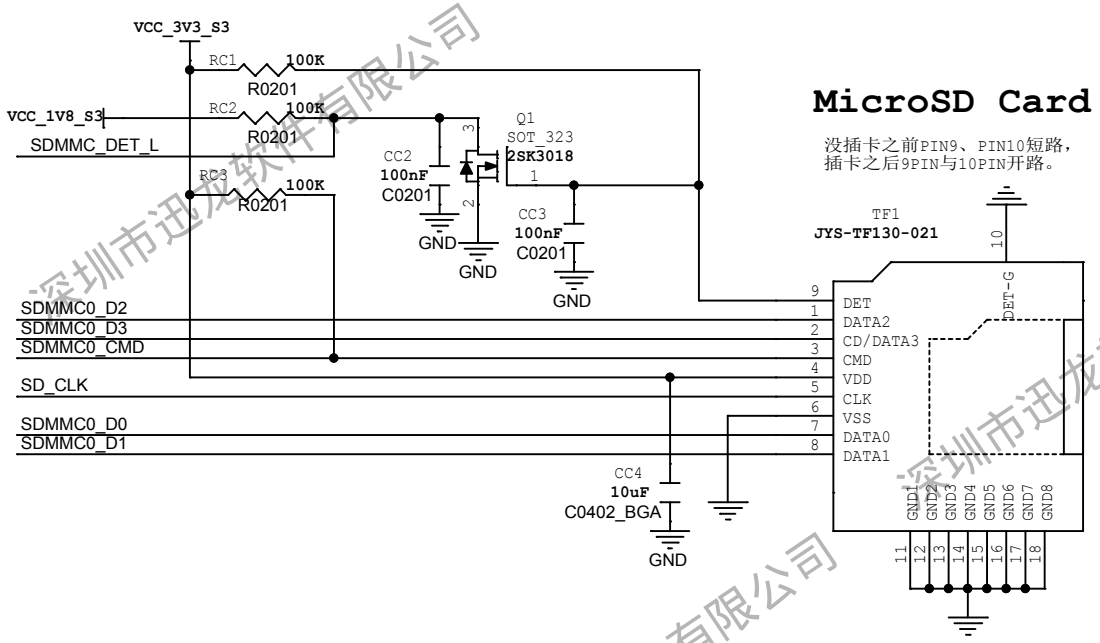
Shenzhen Xunlong Software Co., Ltd					
Project:	OPi 5 Pro				
File:	16.Flash-eMMC Flash				
Date:	Monday, June 03, 2024			Rev:	V1.2
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TF CARD

SDMMC0_D0
SDMMC0_D1
SDMMC0_D2
SDMMC0_D3

SDMMC0_CMD
SD_CLK

SDMMC_DET_L



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Shenzhen Xunlong Software Co., Ltd					
Project:	OPi 5 Pro				
File:	17.Flash-TF Card				
Date:	Monday, June 03, 2024			Rev:	V1.2
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MIPI-CSI0_RX

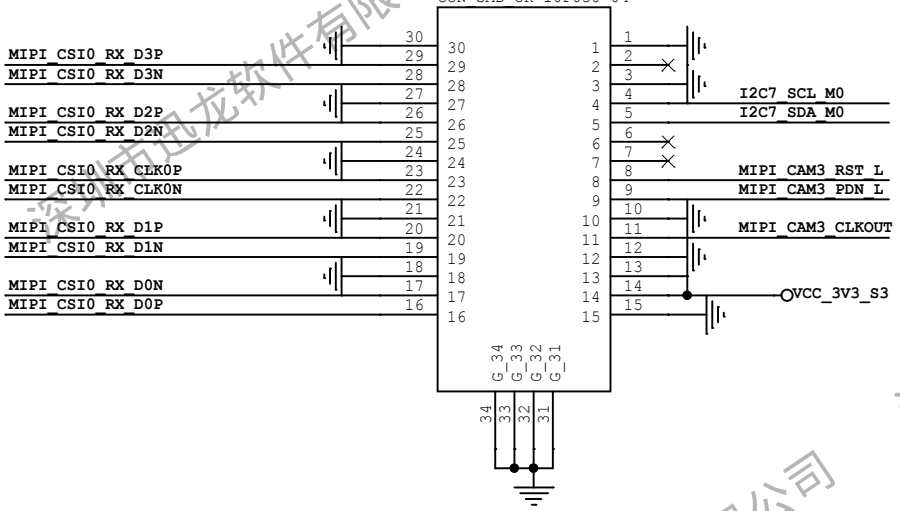
MIPI_CSI0_RX_D0P
MIPI_CSI0_RX_D0N
MIPI_CSI0_RX_D1P
MIPI_CSI0_RX_D1N
MIPI_CSI0_RX_D2P
MIPI_CSI0_RX_D2N
MIPI_CSI0_RX_D3P
MIPI_CSI0_RX_D3N
MIPI_CSI0_RX_CLK0P
MIPI_CSI0_RX_CLK0N

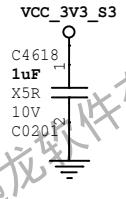
MIPI_CAM3_CLKOUT
MIPI_CAM3_RST_L
MIPI_CAM3_PDN_L

I2C7_SCL_M0
I2C7_SDA_M0

Cam1

OK-10F030-04
CON_SMD_OK-10F030-04





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Shenzhen Xunlong Software Co., Ltd					
Project:	OPi 5 Pro				
File:	18.VI-Cam MIPI CSI0-RX				
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VI-Camera DPHY_RX

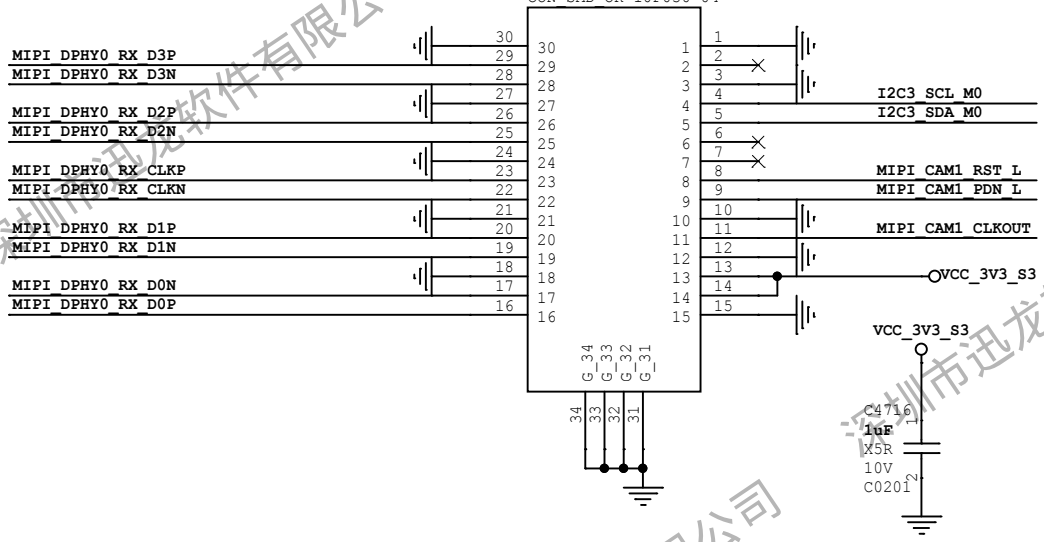
MIPI_DPHY0_RX_CLKP
MIPI_DPHY0_RX_CLKN
MIPI_DPHY0_RX_D0P
MIPI_DPHY0_RX_D0N
MIPI_DPHY0_RX_D1P
MIPI_DPHY0_RX_D1N
MIPI_DPHY0_RX_D2P
MIPI_DPHY0_RX_D2N
MIPI_DPHY0_RX_D3P
MIPI_DPHY0_RX_D3N

MIPI_CAM1_CLKOUT
MIPI_CAM1_PDN_L
MIPI_CAM1_RST_L
I2C3_SCL_M0
I2C3_SDA_M0

MIPI D-PHY0

Cam2

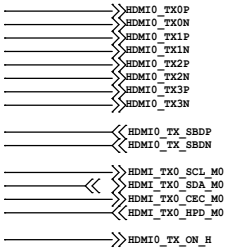
OK-10F030-04
CON_SMD_OK-10F030-04

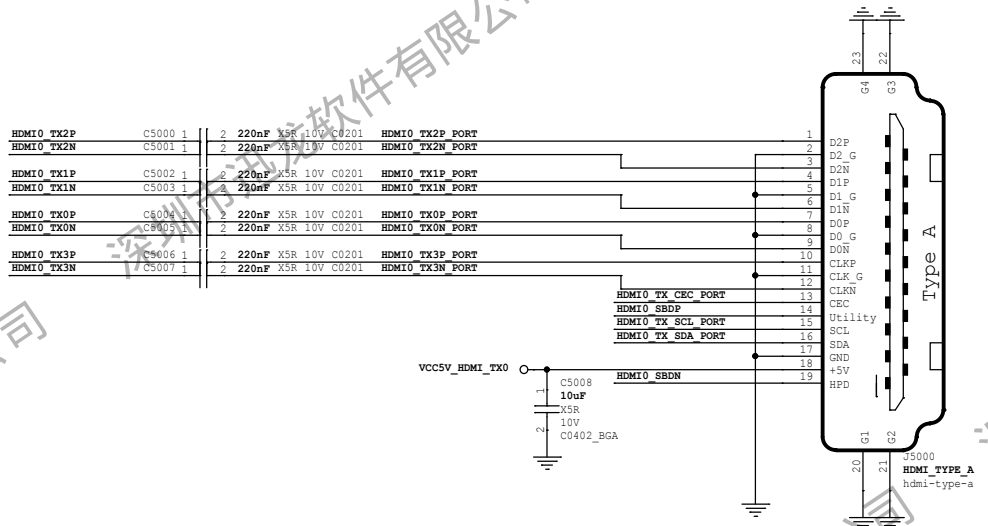


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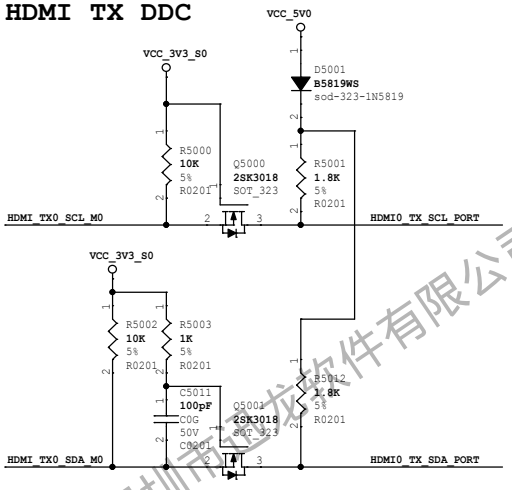
Shenzhen Xunlong Software Co., Ltd					
Project:	OPi 5 Pro				
File:	19.VI-Cam MIPI-DPHY-RX				
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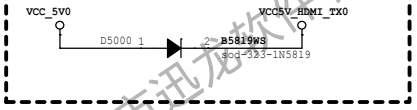
HDMI2.1 TX0

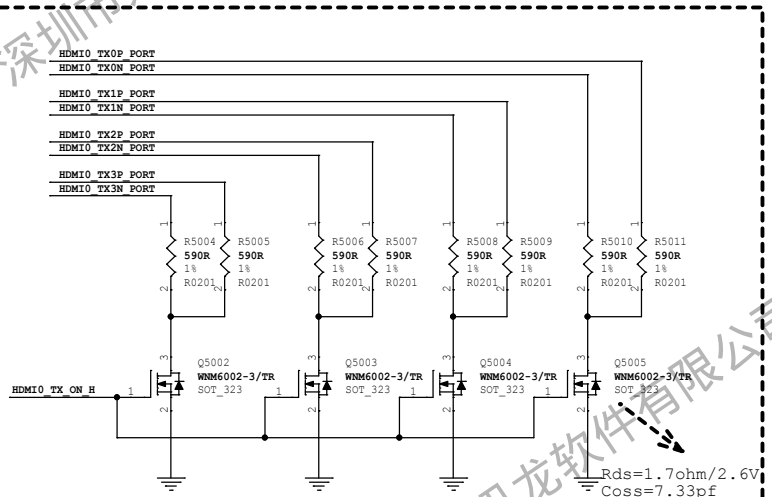




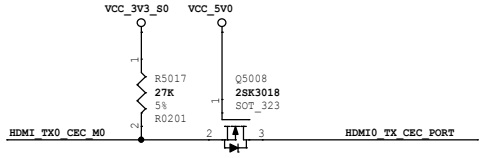
HDMI TX DDC

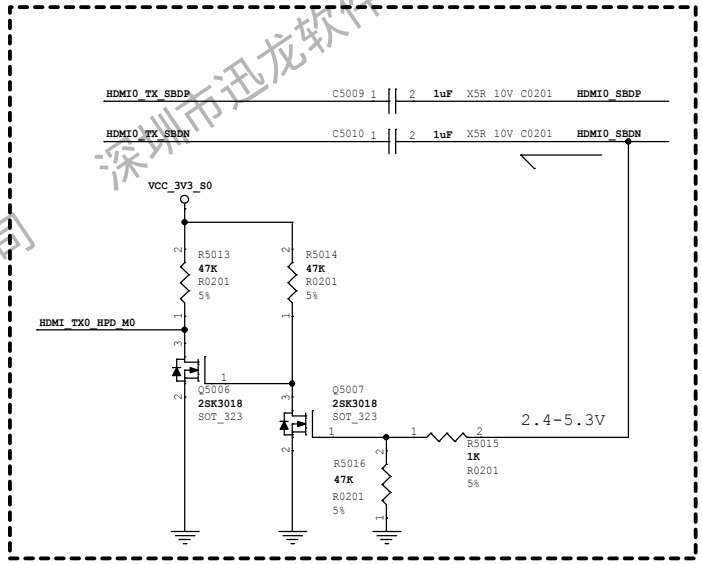






HDMI TX CEC

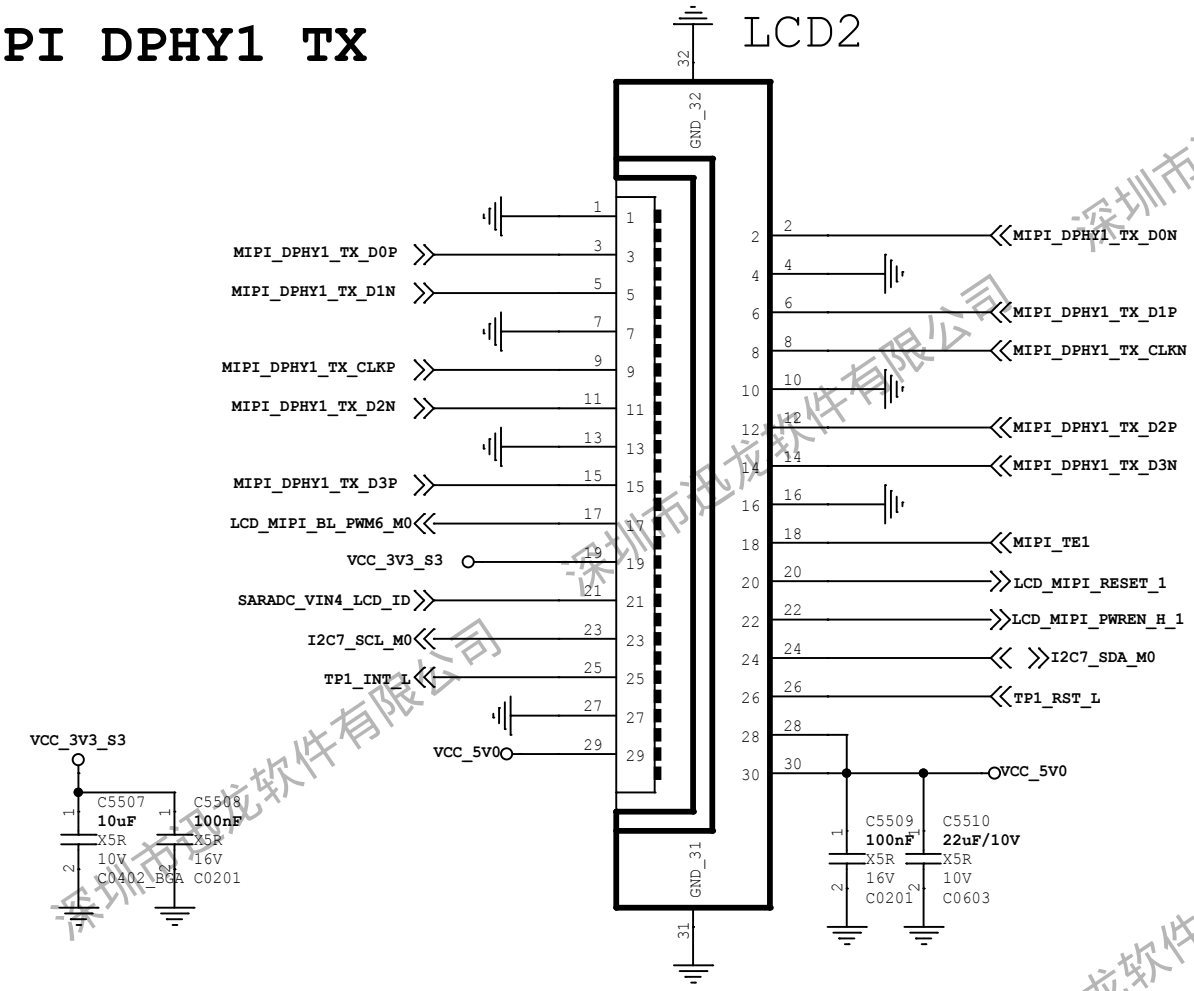




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Shenzhen Xunlong Software Co., Ltd					
Project:	OPi 5 Pro				
File:	20.VO-HDMI2.1 TX				
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MIPI DPHY1 TX

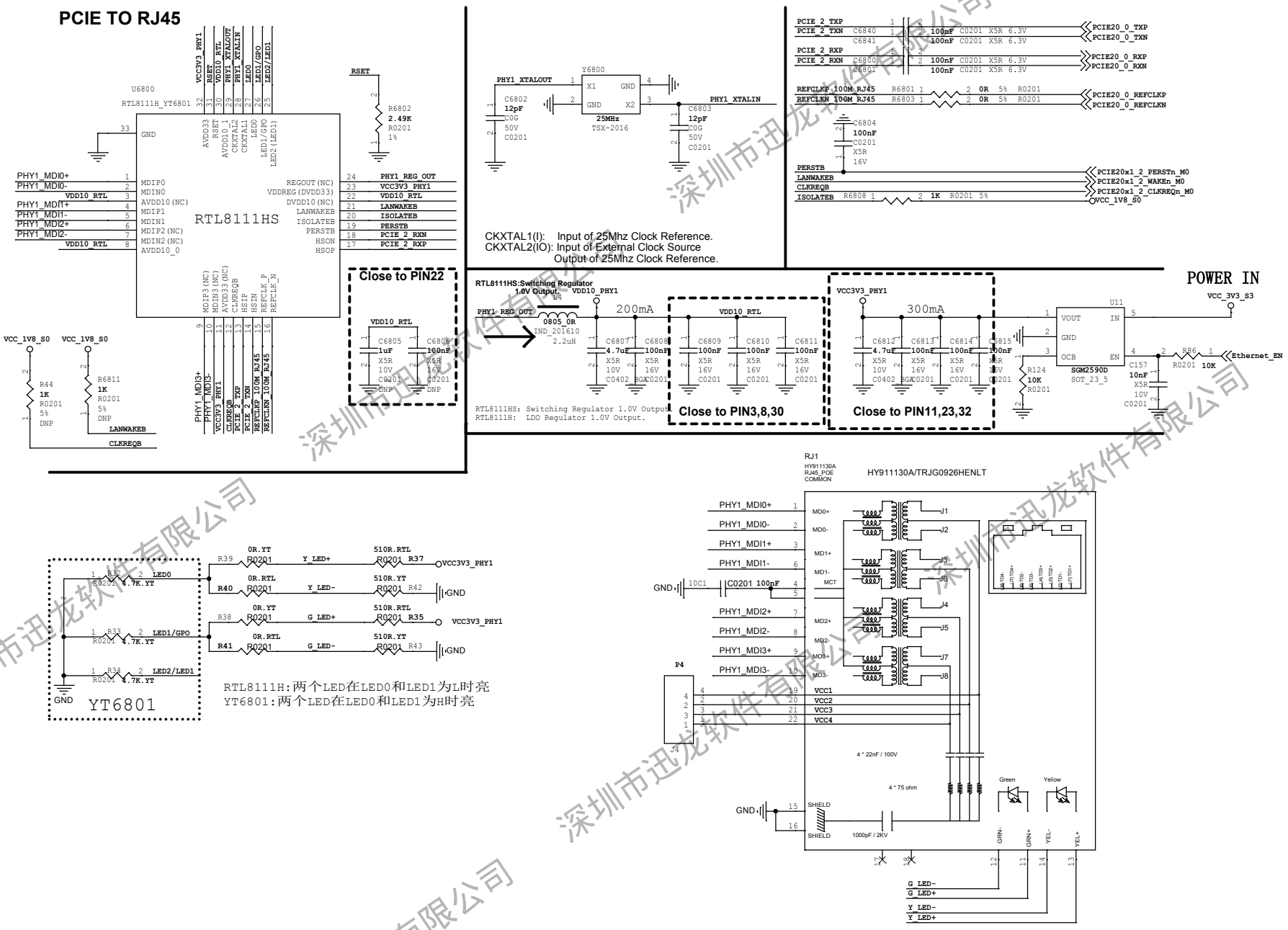


FPC Pin List

Pin1 :GND
Pin2 :D0N
Pin3 :D0P
Pin4 :GND
Pin5 :D1N
Pin6 :D1P
Pin7 :GND
Pin8 :CLKN/AUXN
Pin9 :CLKP/AUXP
Pin10:GND
Pin11:D2N
Pin12:D2P
Pin13:GND
Pin14:D3N
Pin15:D3P
Pin16:GND
Pin17:LCD_PWM_BL
Pin18:LCD_TE
Pin19:VCC3V3_LCD
Pin20:LCD_RST
Pin21:LCD_ID
Pin22:LCD_PWREN
Pin23:TP_I2C_SCL
Pin24:TP_I2C_SDA
Pin25:TP_INT
Pin26:TP_RST
Pin27:GND
Pin28:5V0
Pin29:5V0
Pin30:5V0

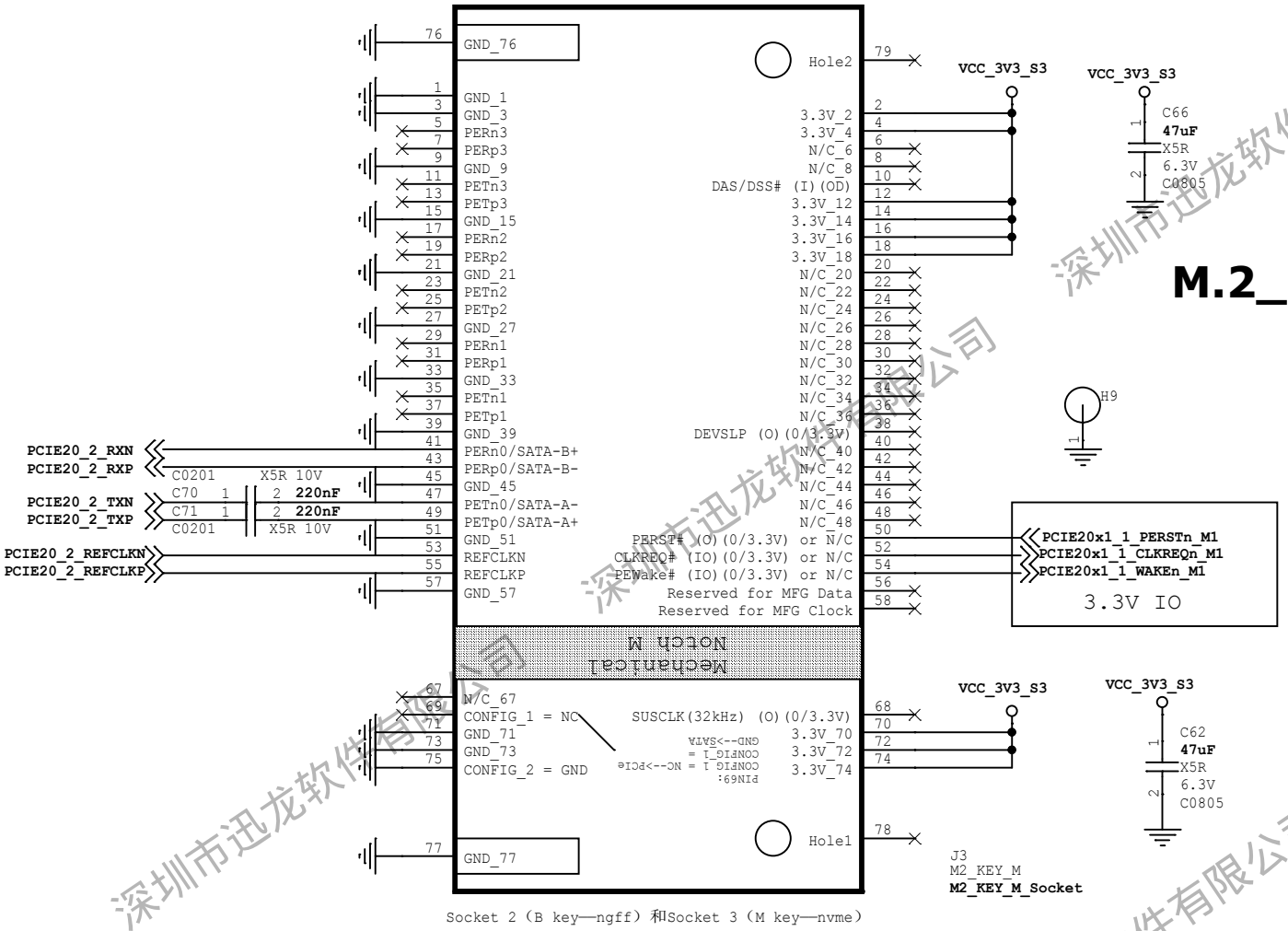
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Shenzhen Xunlong Software Co., Ltd					
Project:	OPi 5 Pro				
File:	21.VO-MIPI_LCD				
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Shenzhen Xunlong Software Co., Ltd					
Project:	OPi 5 Pro				
File:	22.Ethernet-PCIE				
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Socket 2（B key—ngff）和Socket 3（M key—nvme）

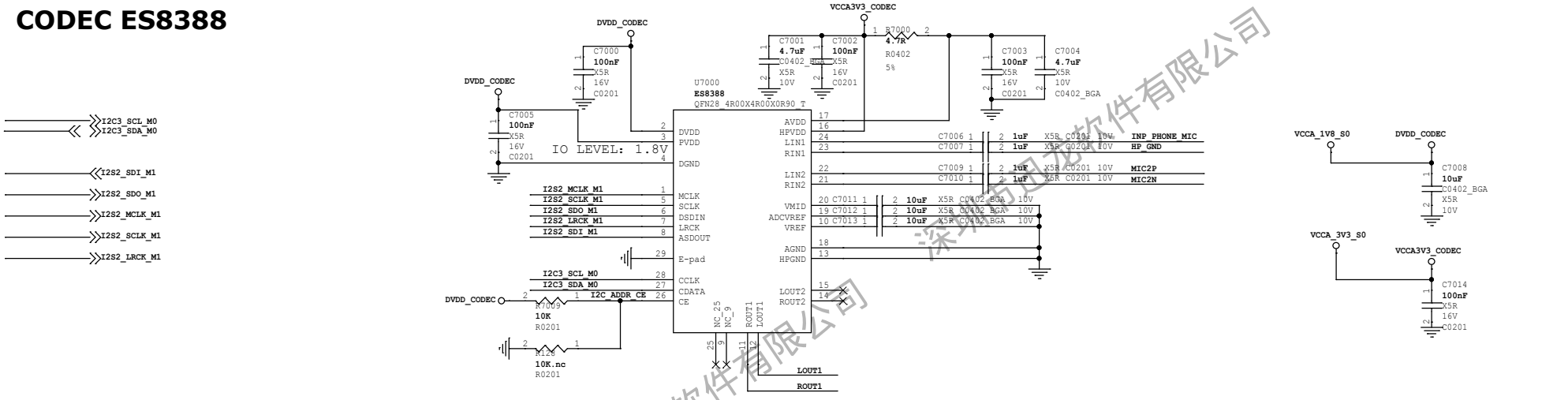
M.2 NVME

M.2_PCIe/SATA_ Interface

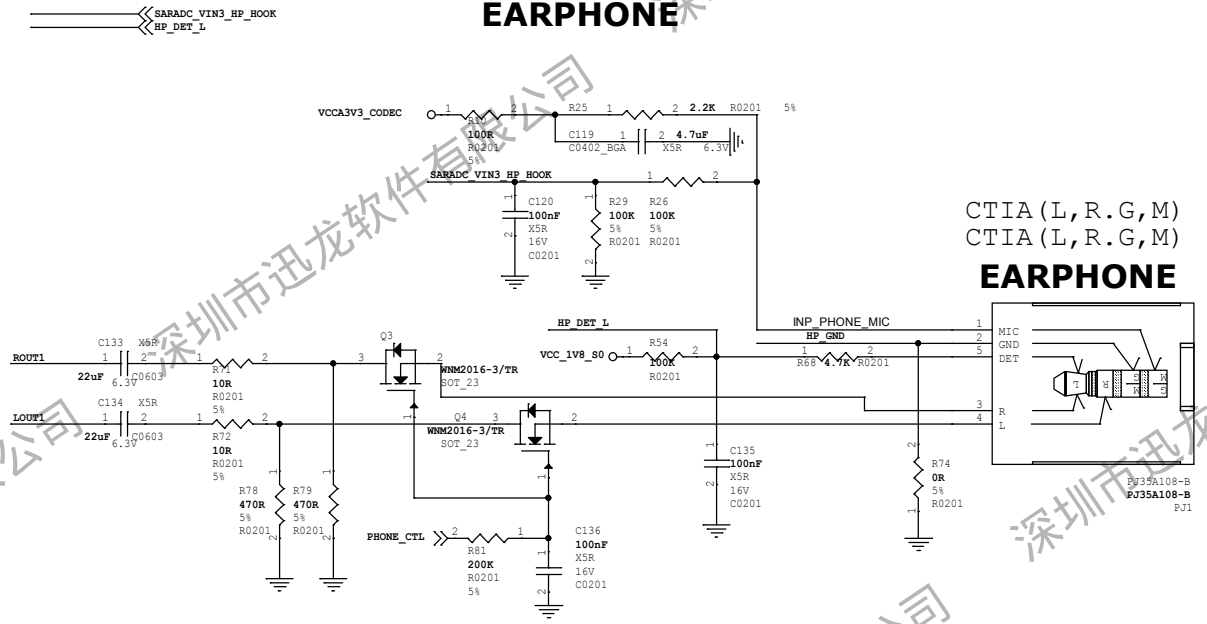
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Shenzhen Xunlong Software Co., Ltd					
Project:	OPi 5 Pro				
File:	23.M.2_PCIe_SATA				
Date:	Monday, June 03, 2024			Rev:	V1.2
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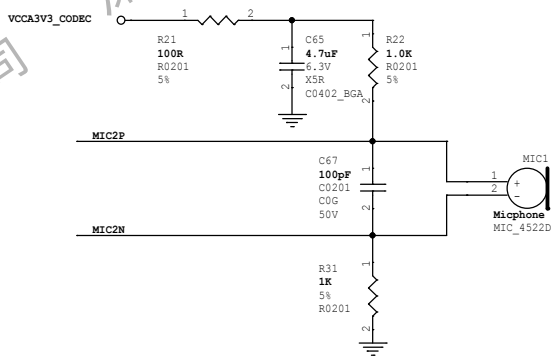
CODEC ES8388



EARPHONE



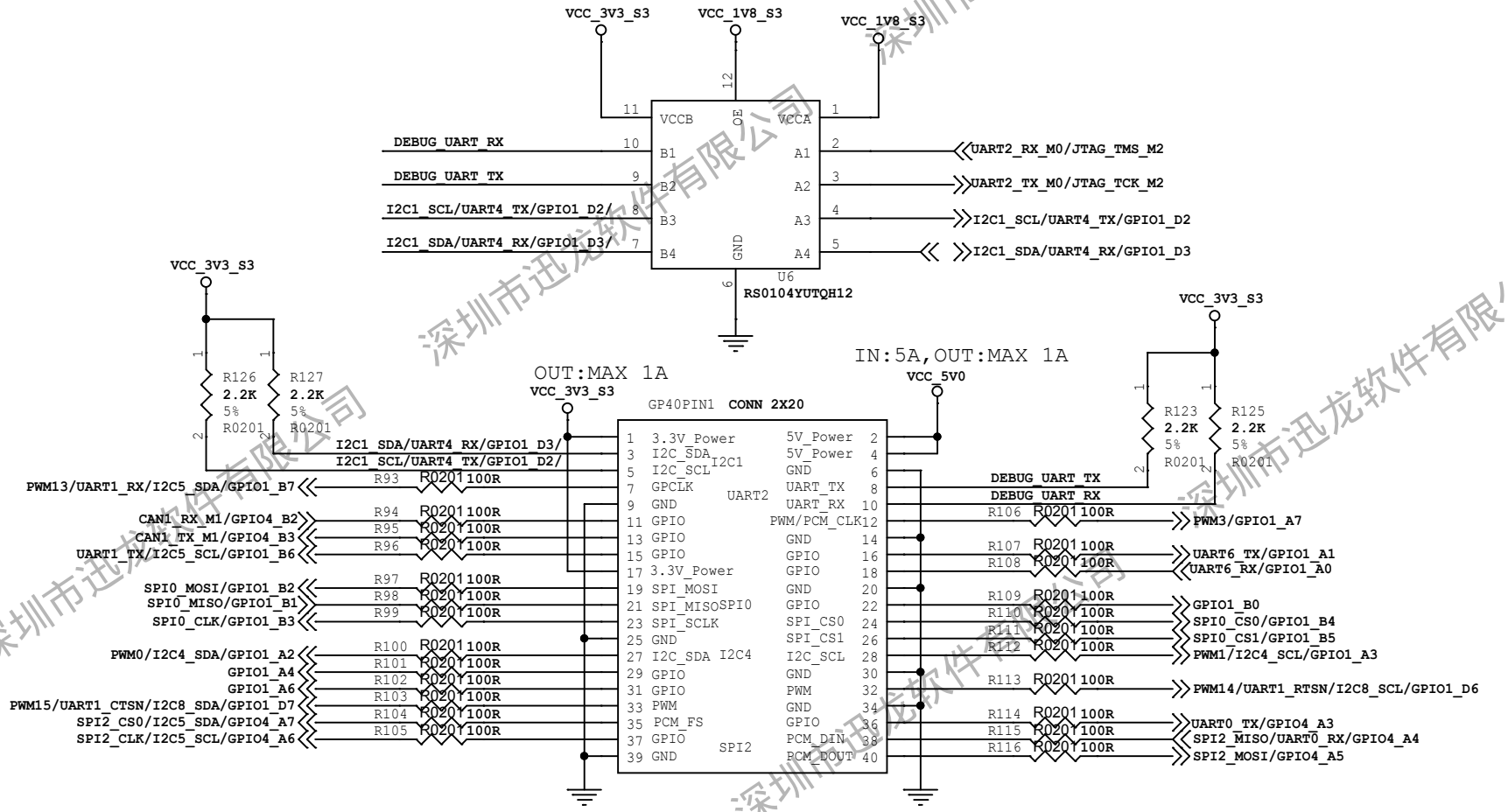
Analog MIC



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Shenzhen Xunlong Software Co., Ltd					
Project:	OPi 5 Pro				
File:	24.Audio Codec(ES8388)				
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EXT I/O



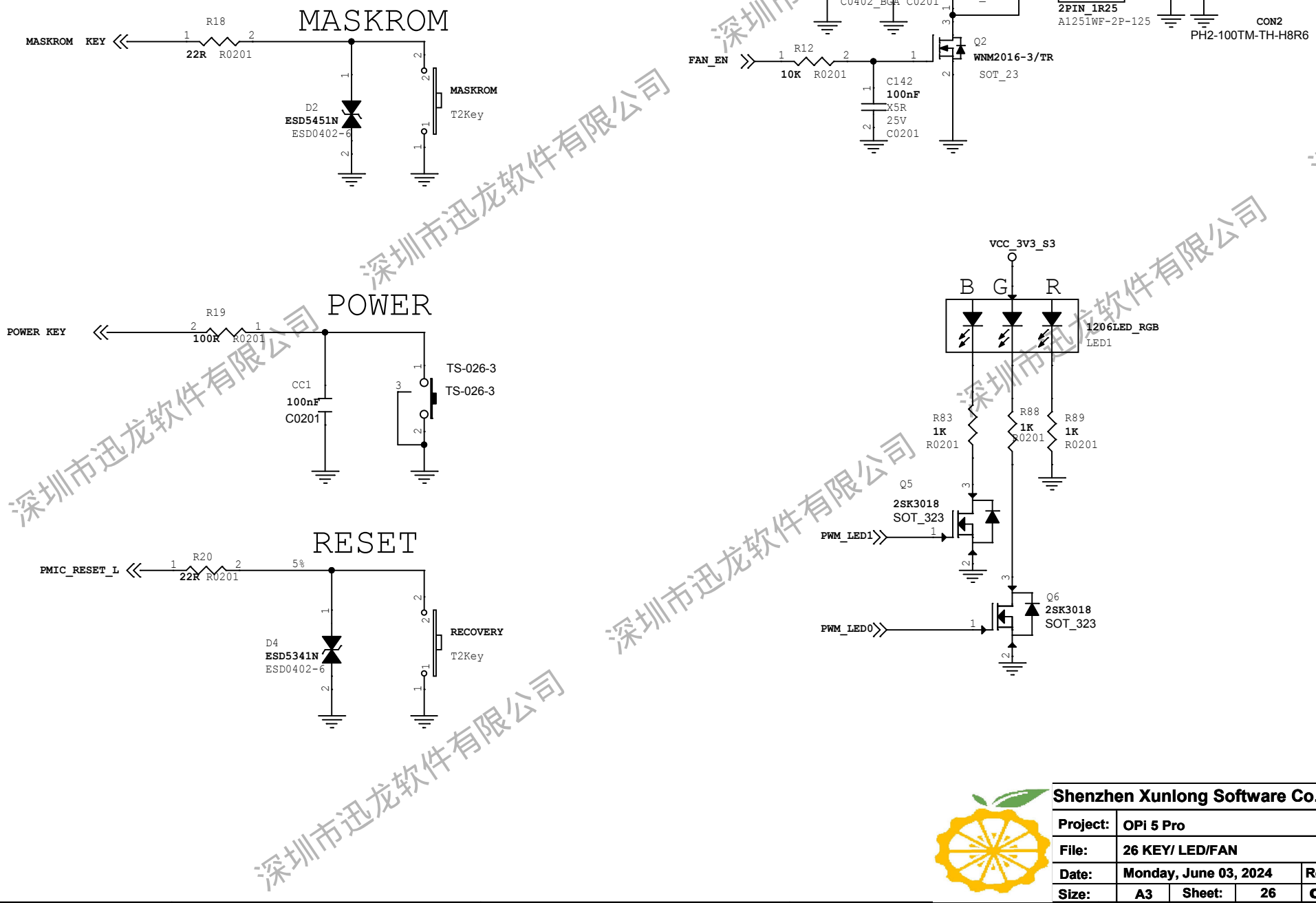
HOLE_500CIR300 H5

HOLE_500CIR300 H6

H4 H3 H2 H1

Shenzhen Xunlong Software Co., Ltd					
Project:	OPi 5 Pro				
File:	25.EXT_ GPIO/Debug				
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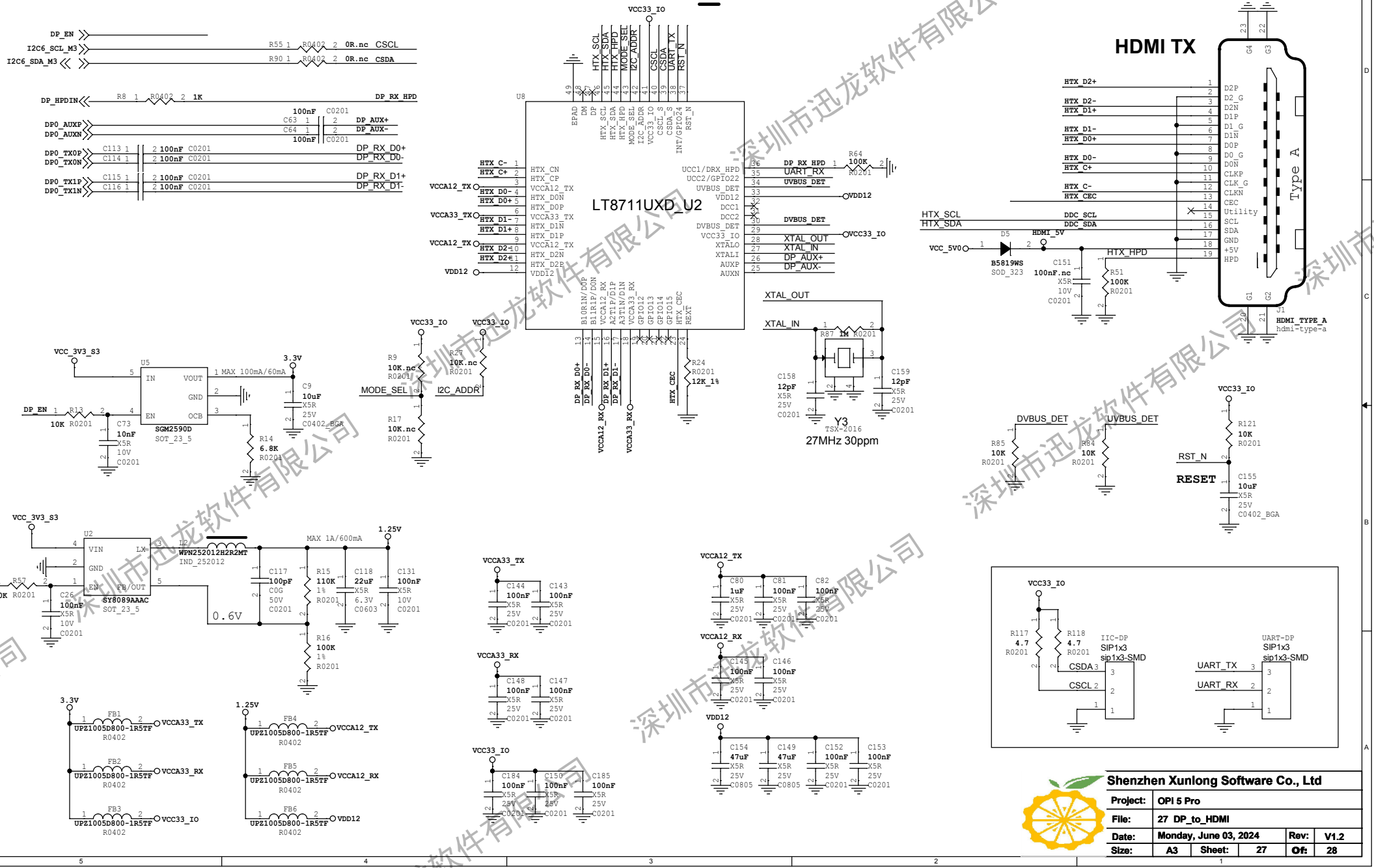
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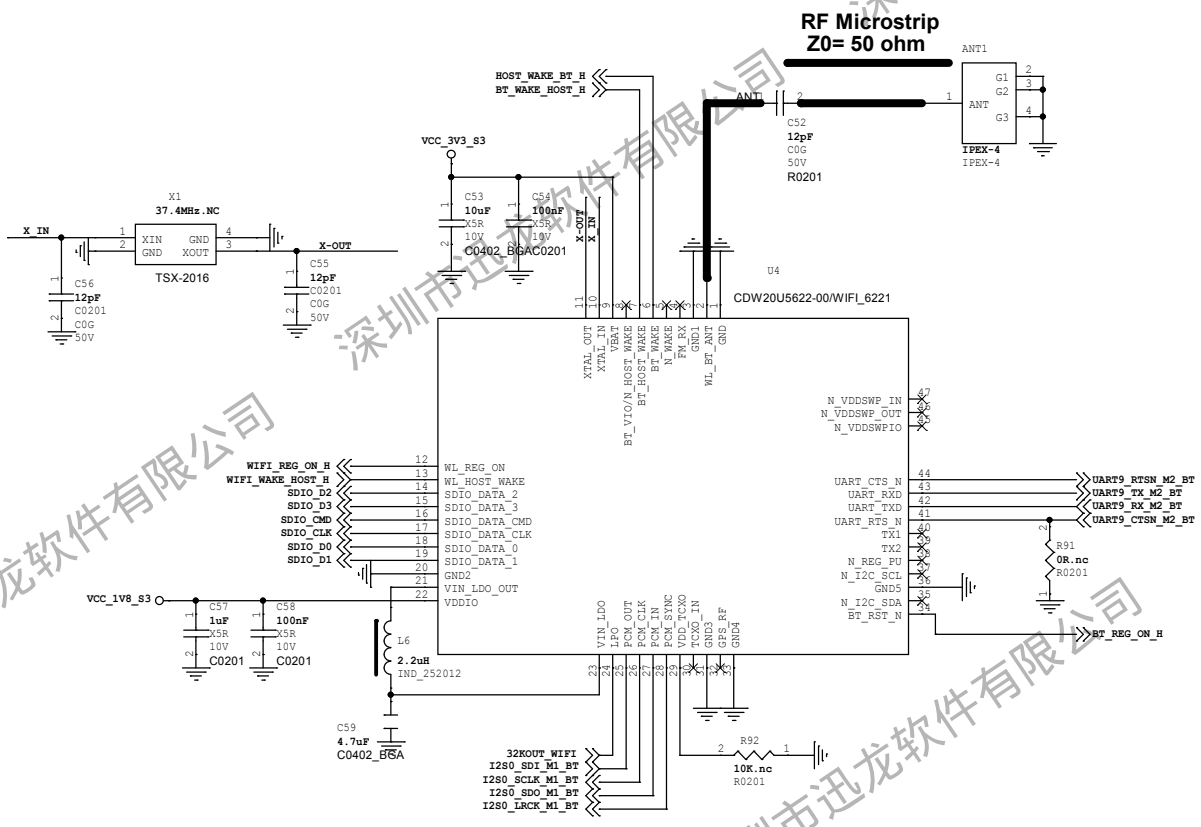
Shenzhen Xunlong Software Co., Ltd					
Project:	OPi 5 Pro				
File:	26 KEY/ LED/FAN				
Date:	Monday, June 03, 2024			Rev:	V1.2
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LT8711UXD_U2



Shenzhen Xunlong Software Co., Ltd					
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File:	27 DP_to_HDMI				
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Shenzhen Xunlong Software Co., Ltd					
Project:	OPi 5 Pro				
File:	28.WIFI/BT-SDIO				
Date:	Monday, June 03, 2024			Rev:	V1.2
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